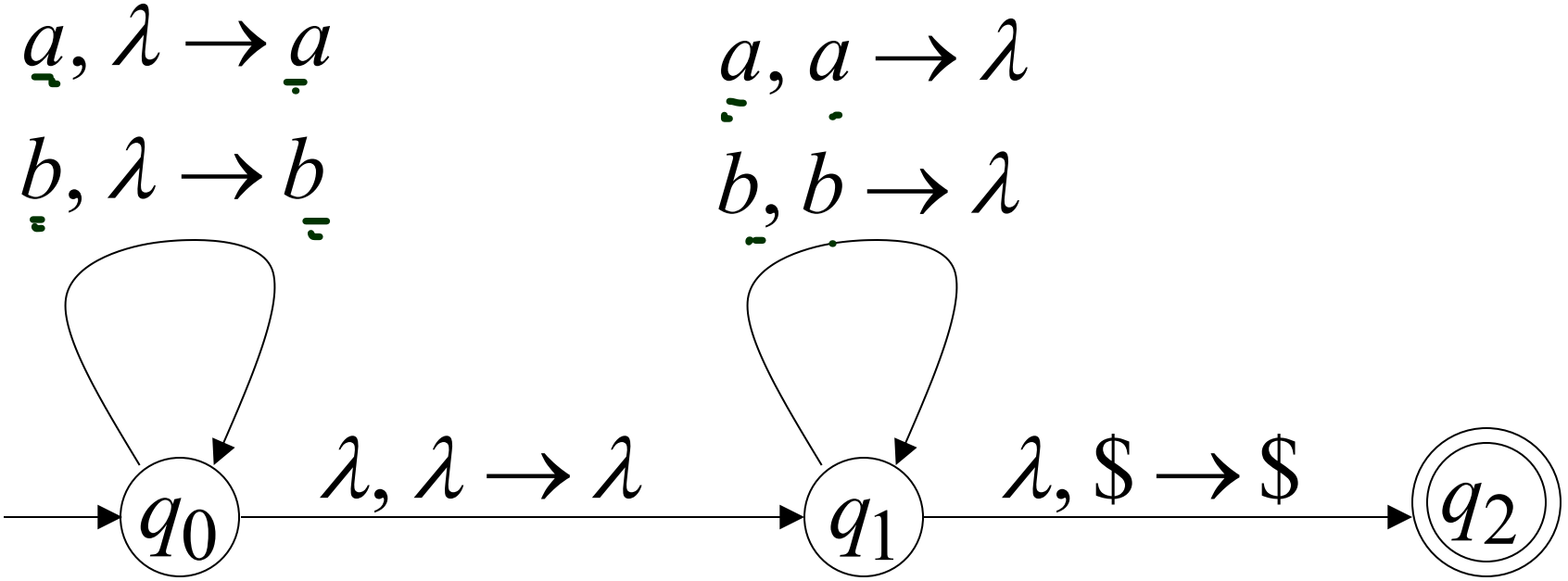
Another PDA example

$$L(M) = \{vv^R : v \in \{a,b\}^*\}$$

PDA M



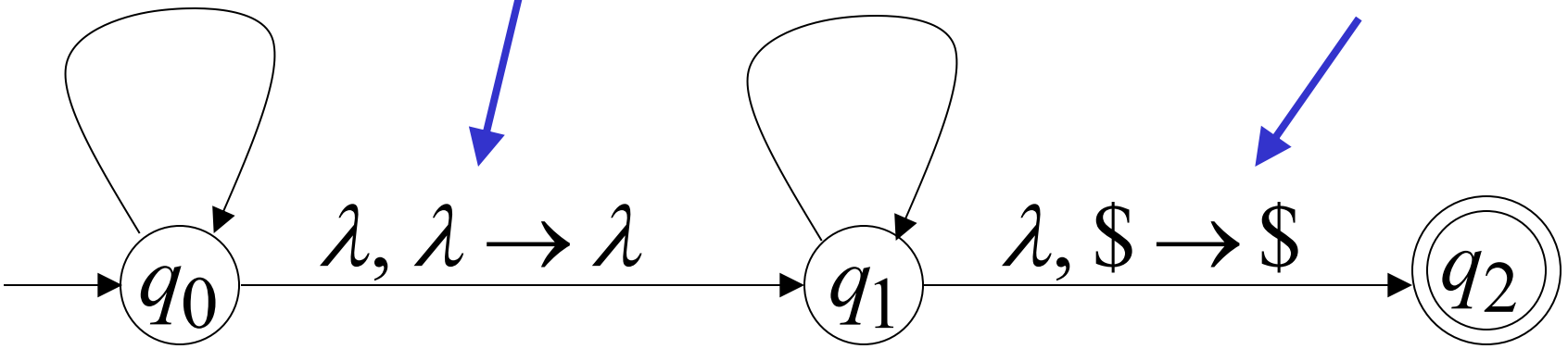
Basic Idea: $L(M) = \{vv^R : v \in \{a,b\}^*\}$

1. Push v on stack

2. Guess middle of input

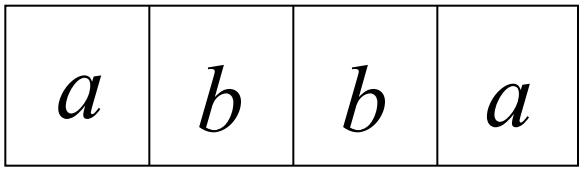
3. Match v^R on input with v on stack

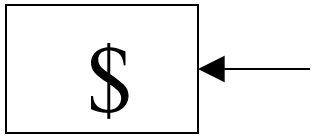
4. Match found



Execution Example: Time 0

Input





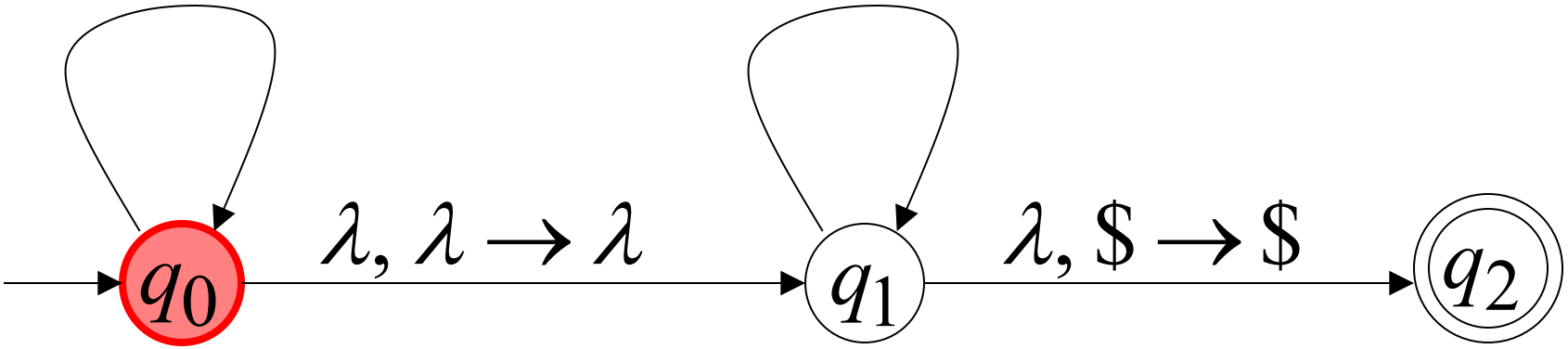
Stack

$a, \lambda \rightarrow a$

$b, \lambda \rightarrow b$

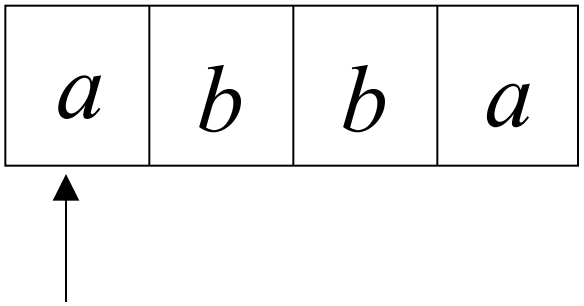
$a, a \rightarrow \lambda$

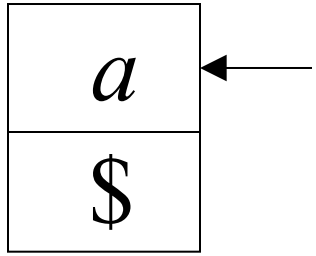
$b, b \rightarrow \lambda$



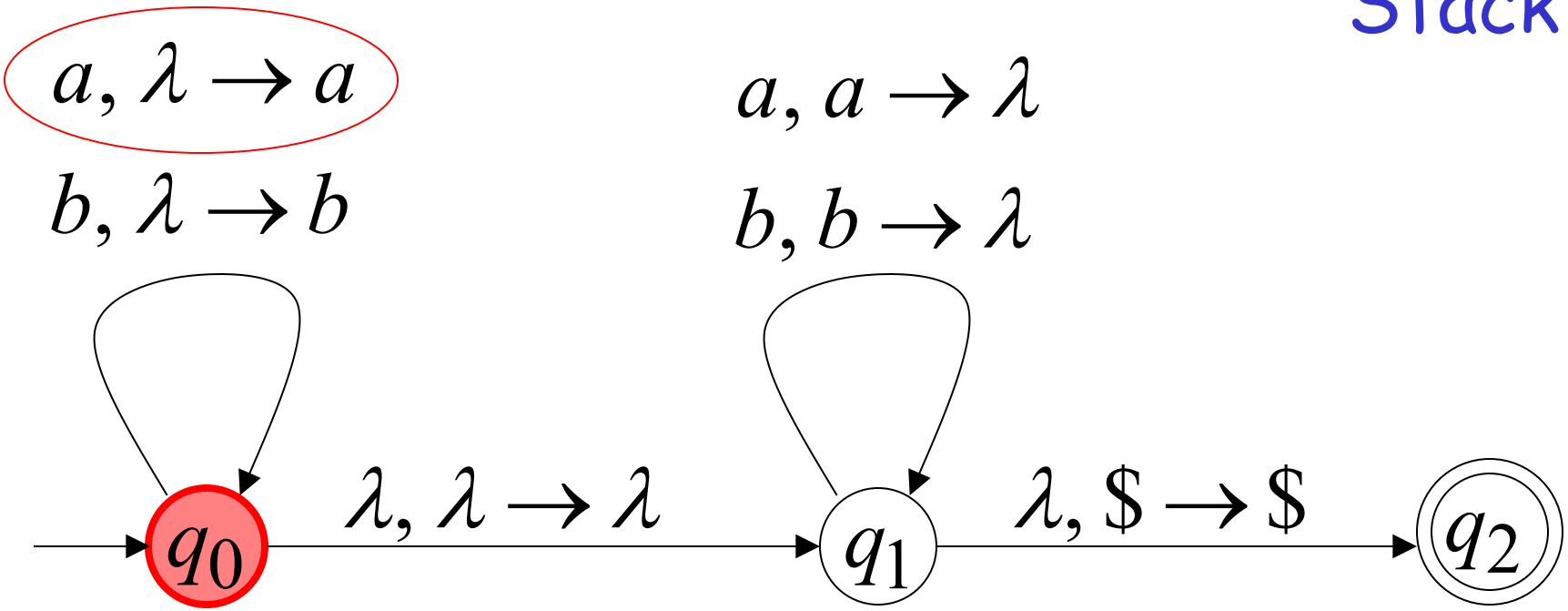
Time 1

Input



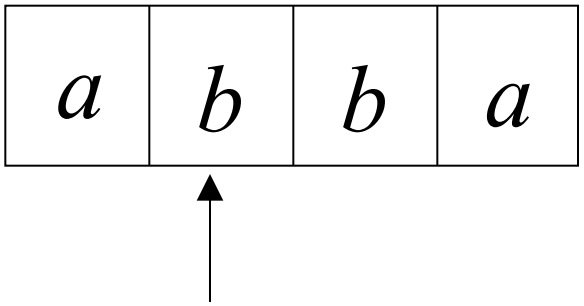


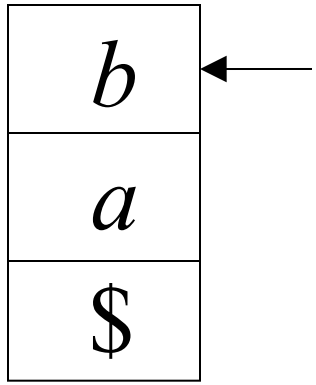
Stack



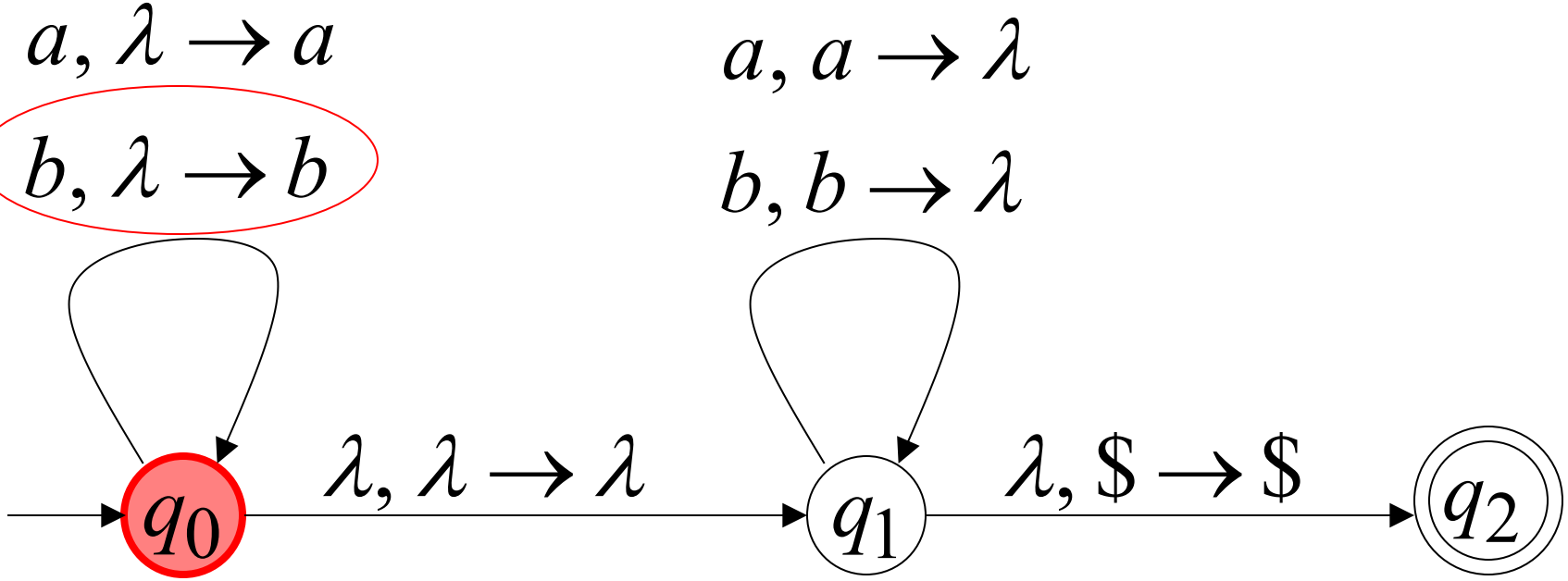
Time 2

Input



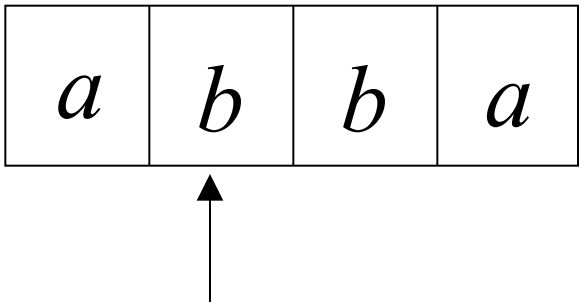


Stack

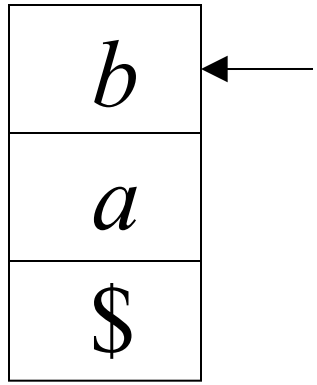


Time 3

Input



Guess the middle of string



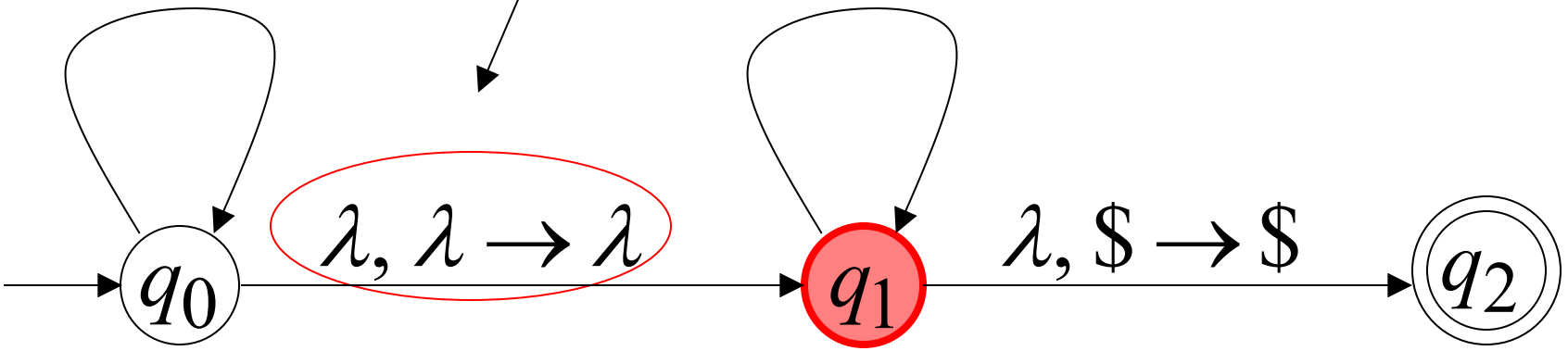
Stack

$a, \lambda \to a$

$b, \lambda \to b$

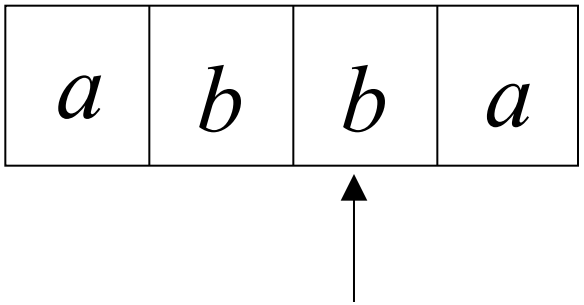
$a, a \to \lambda$

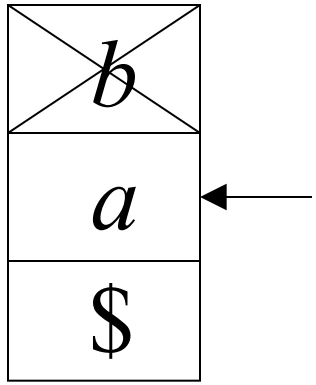
$b, b \to \lambda$



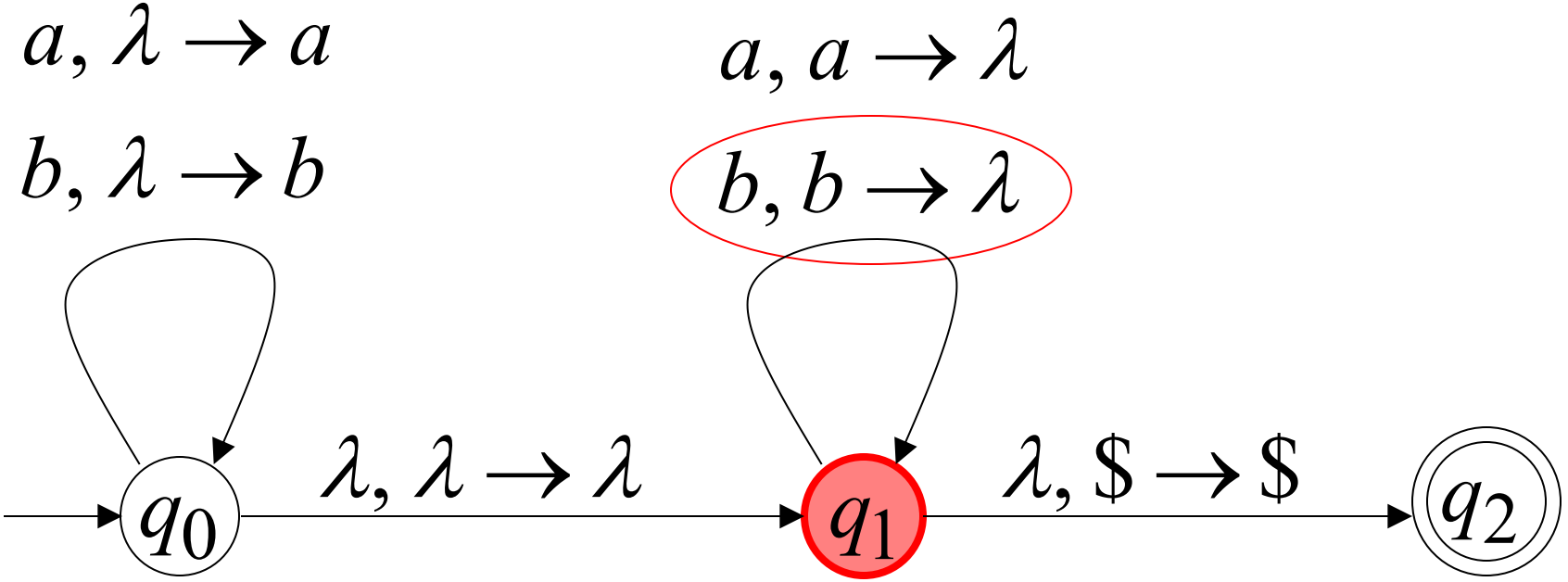
Time 4

Input



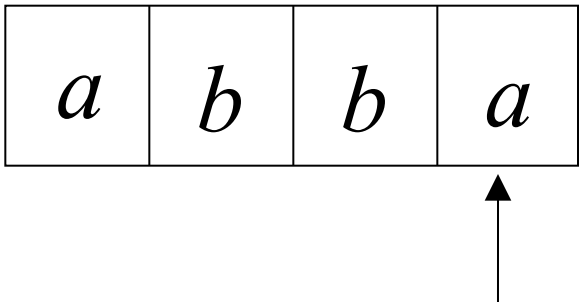


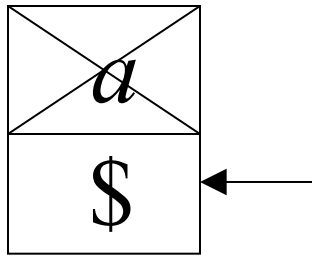
Stack



Time 5

Input





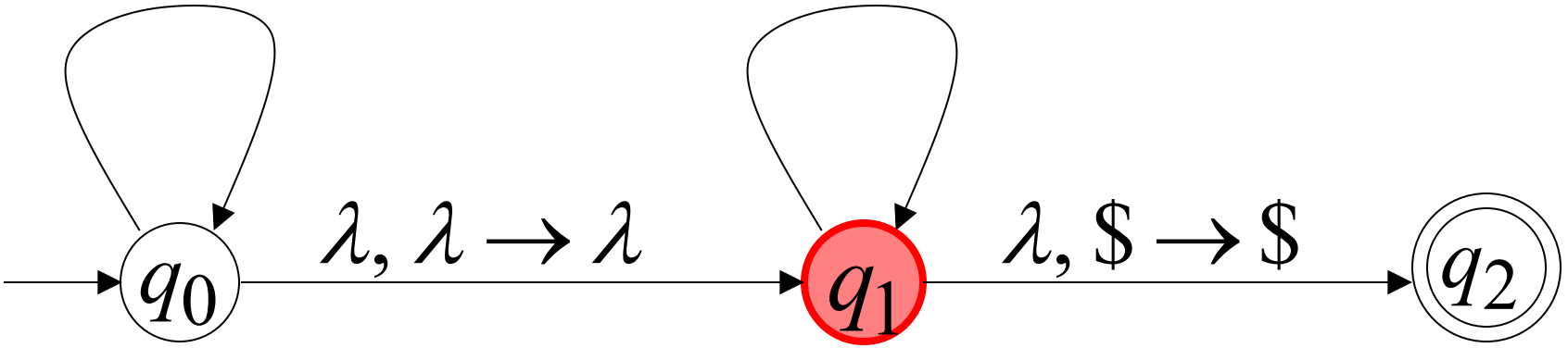
Stack

$a, \lambda \to a$

$b, \lambda \to b$

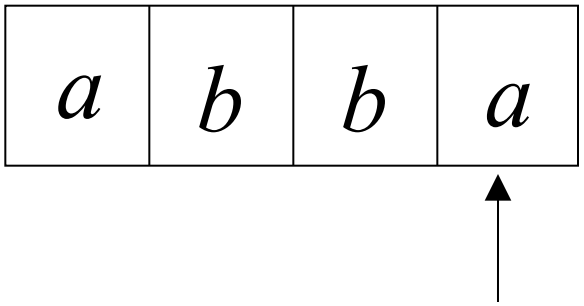
$a, a \to \lambda$

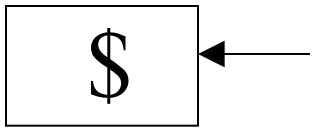
$b, b \to \lambda$



Time 6

Input





Stack

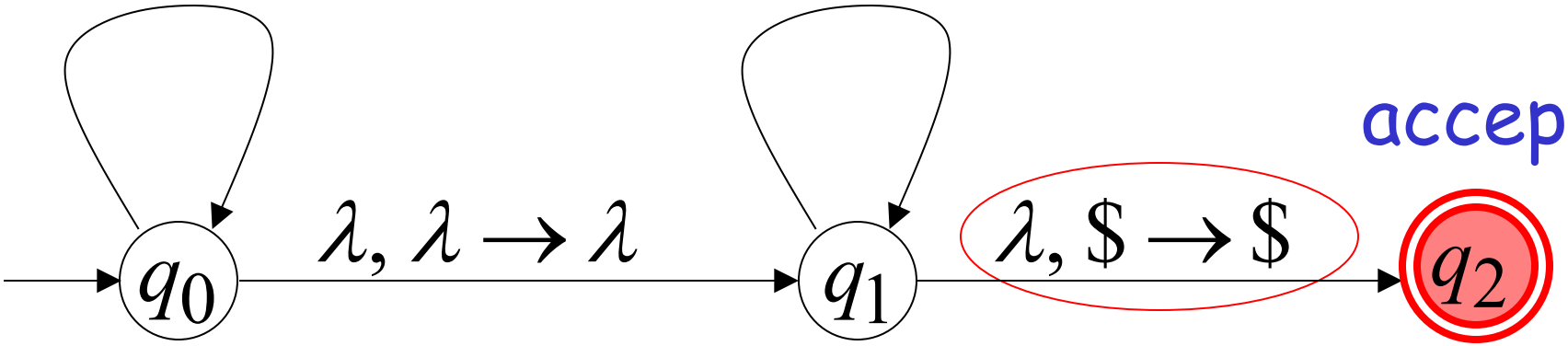
$a, \lambda \rightarrow a$

$b, \lambda \rightarrow b$

$a, a \rightarrow \lambda$

$b, b \rightarrow \lambda$

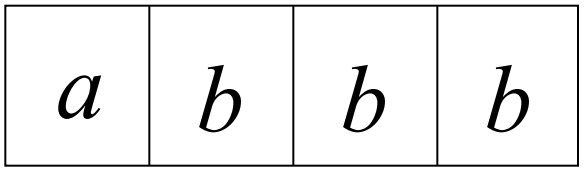
accept

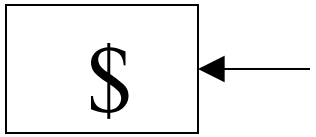


Rejection Example:

Time 0

Input





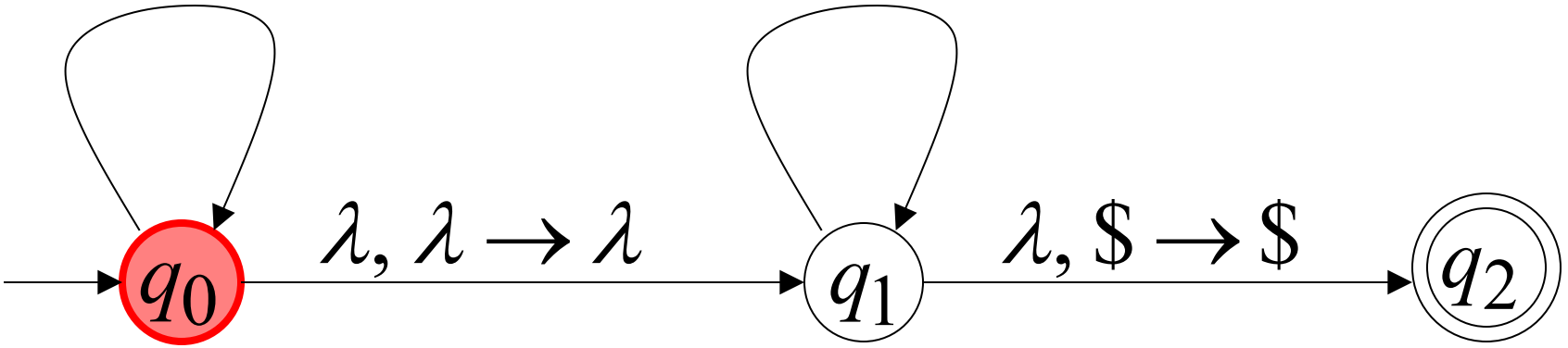
Stack

$a, \lambda \to a$

$b, \lambda \to b$

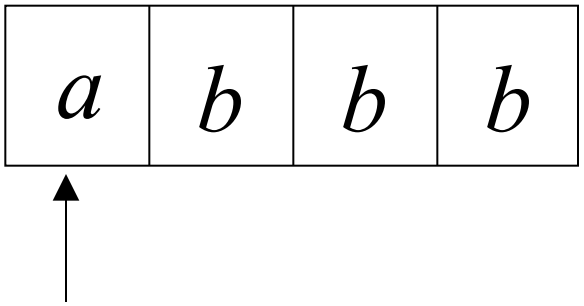
$a, a \to \lambda$

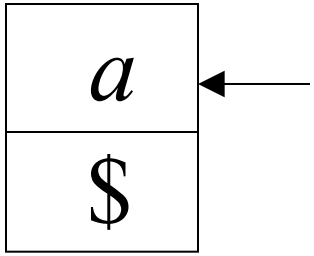
$b, b \to \lambda$



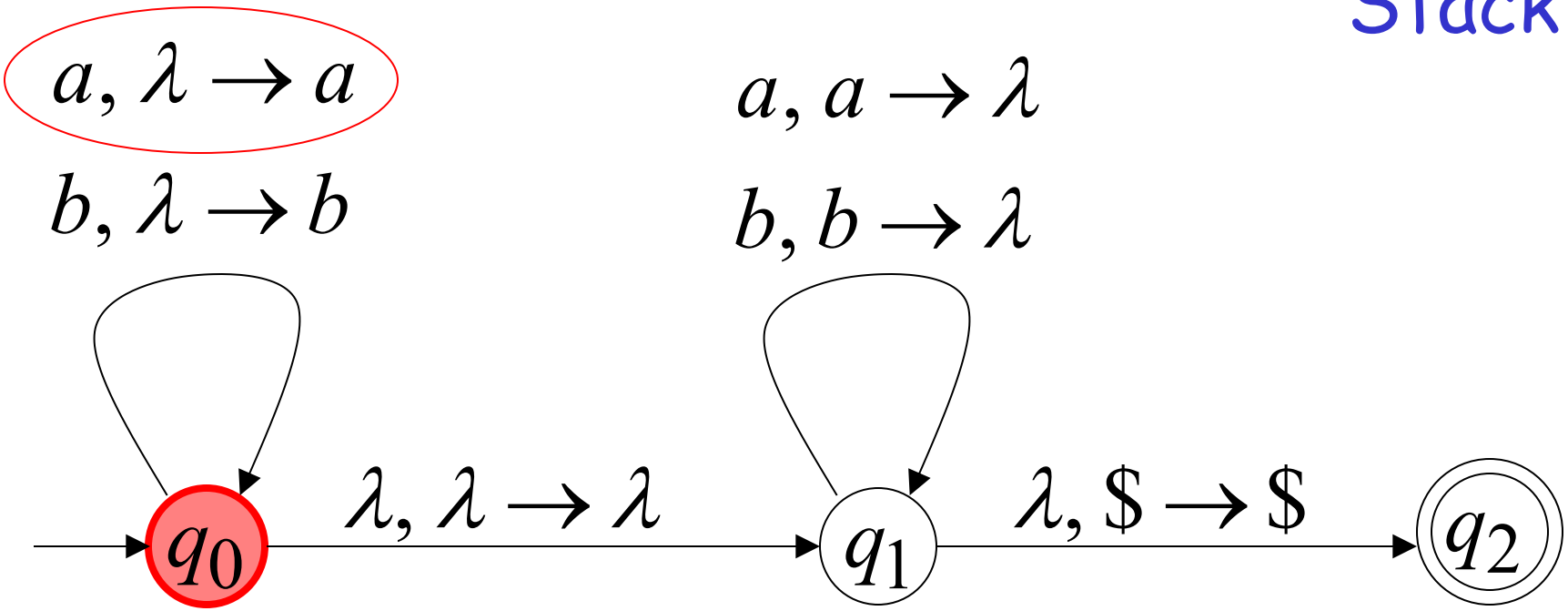
Time 1

Input



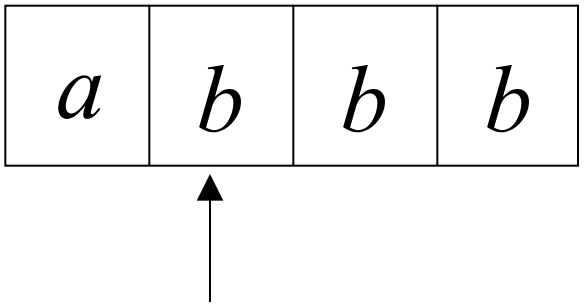


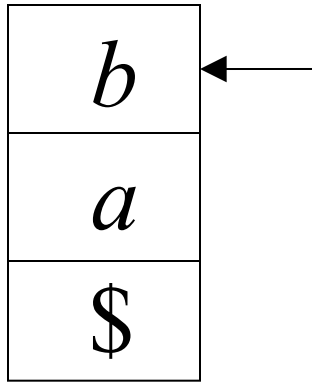
Stack



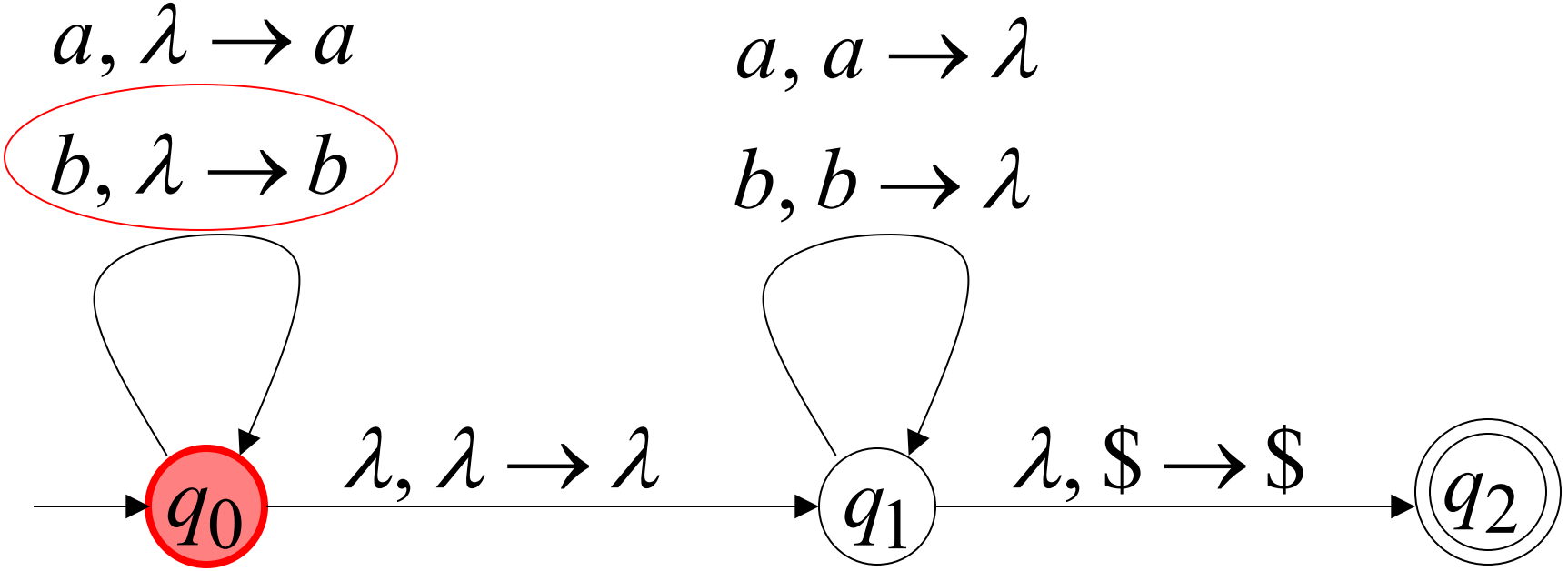
Time 2

Input





Stack



Time 3

Input

a	b	b	b

Guess the middle of string

b
a
\$

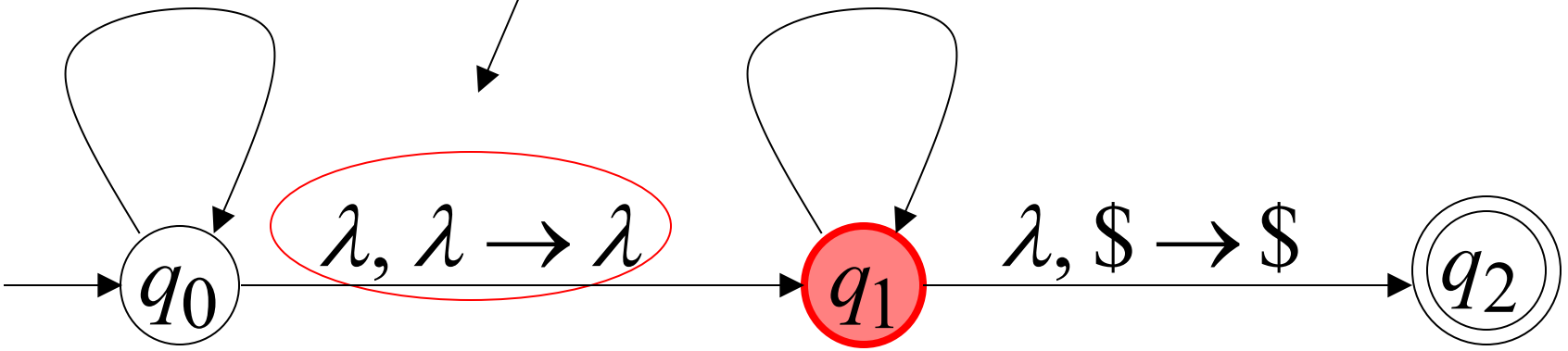
Stack

$a, \lambda \to a$

$b, \lambda \to b$

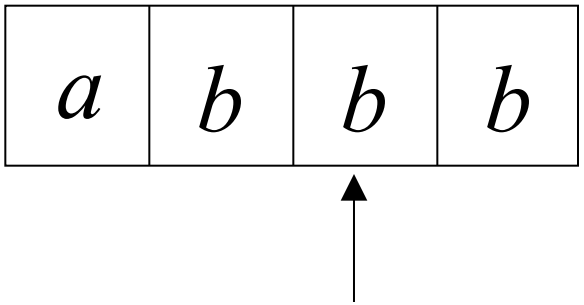
$a, a \to \lambda$

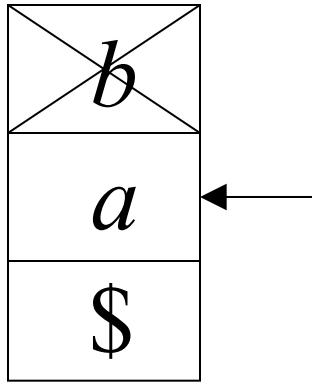
$b, b \to \lambda$



Time 4

Input





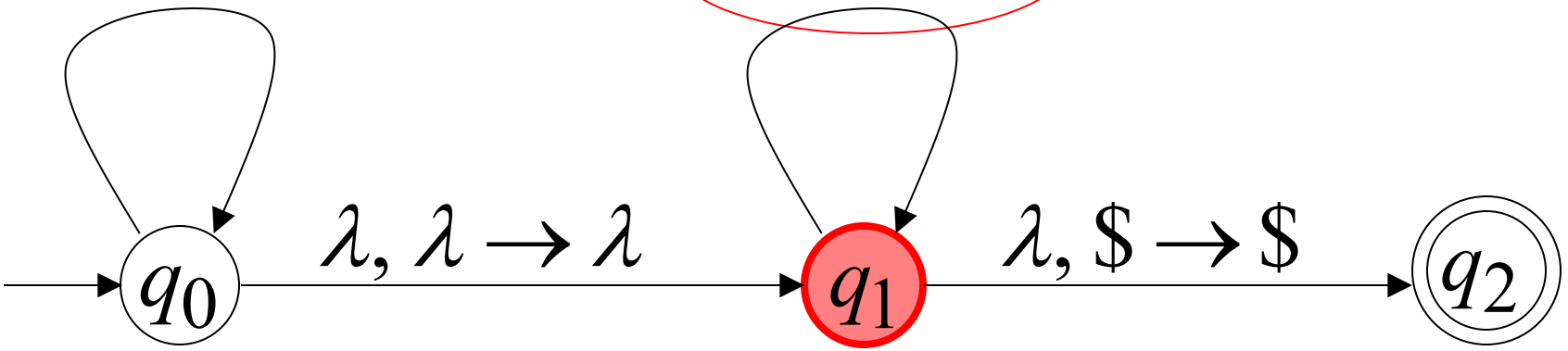
Stack

$a, \lambda \rightarrow a$

$b, \lambda \rightarrow b$

$a, a \rightarrow \lambda$

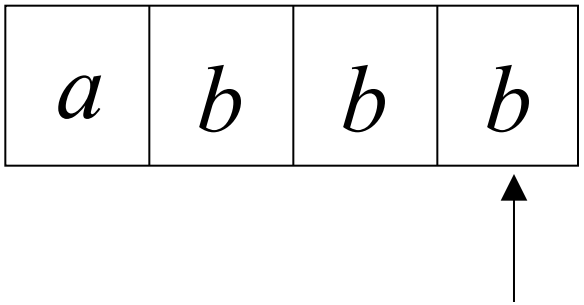
$b, b \rightarrow \lambda$



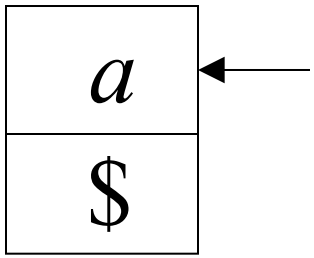
Time 5

There is no possible transition.

Input



Input is not consumed



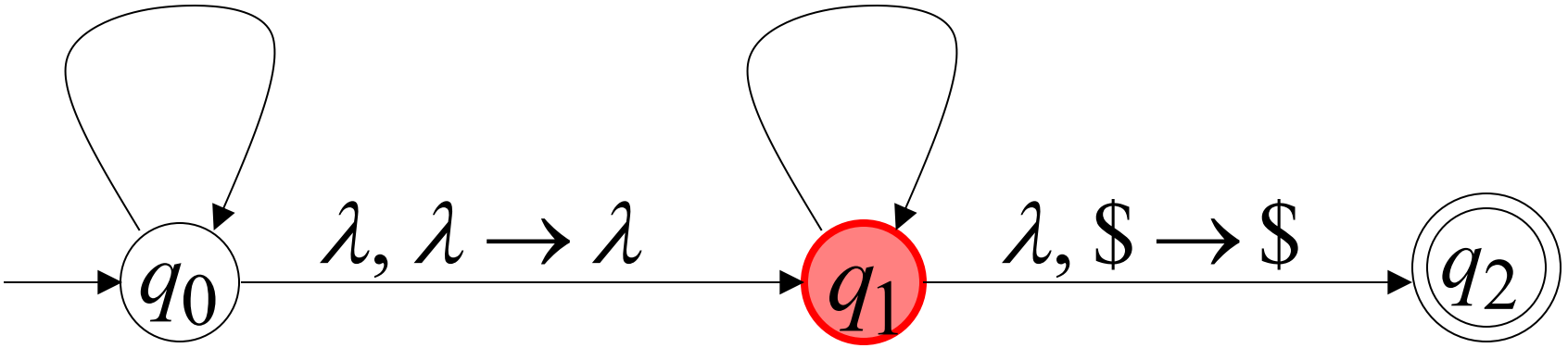
Stack

$a, \lambda \rightarrow a$

$b, \lambda \rightarrow b$

$a, a \rightarrow \lambda$

$b, b \rightarrow \lambda$



Another computation on same string:

Input

Time 0

a	b	b	b

$\$$

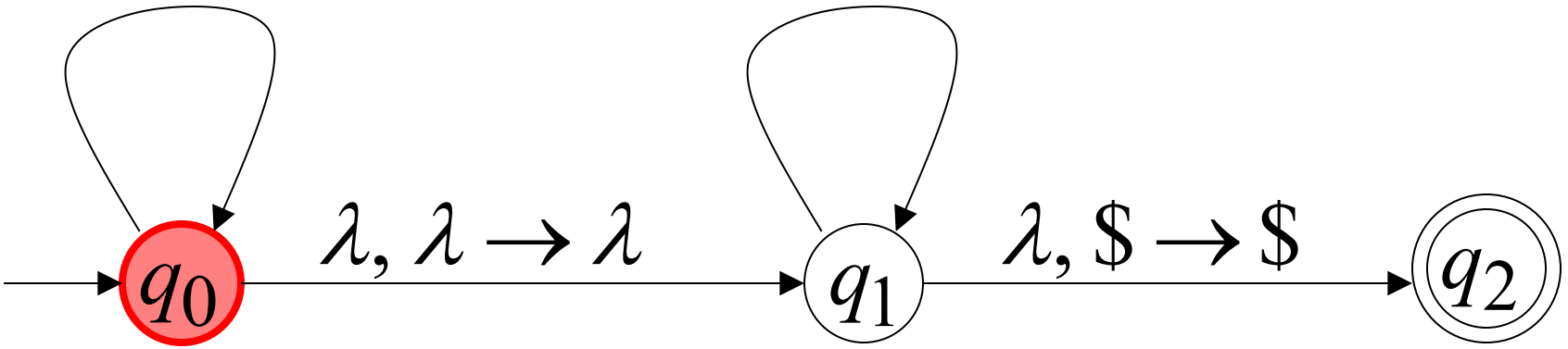
Stack

$a, \lambda \to a$

$b, \lambda \to b$

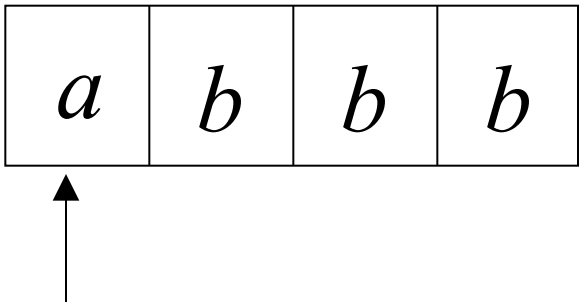
$a, a \to \lambda$

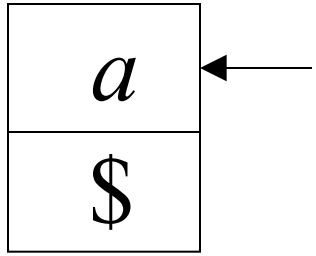
$b, b \to \lambda$



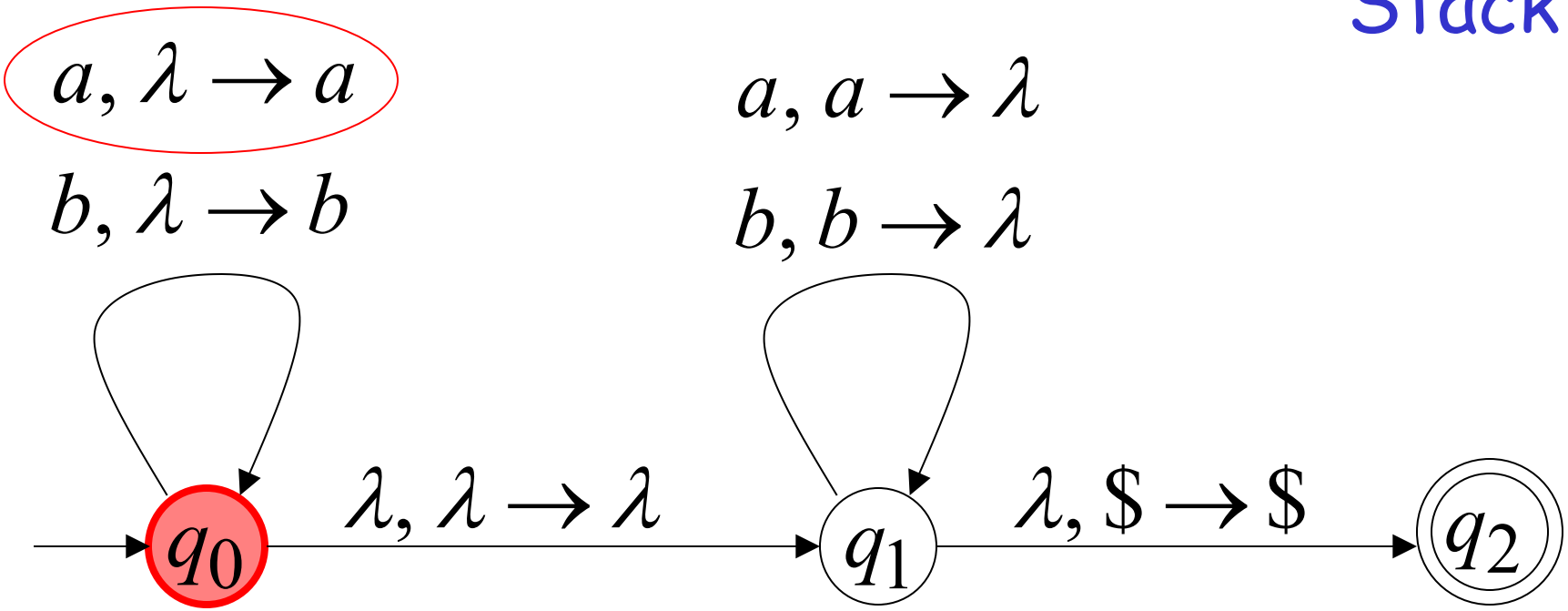
Time 1

Input



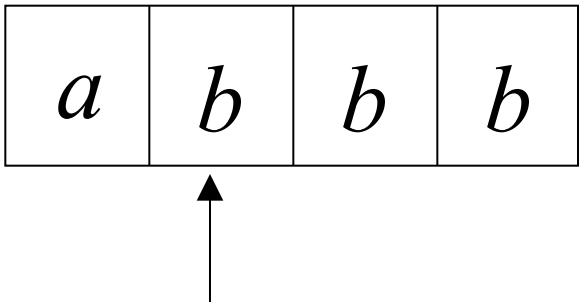


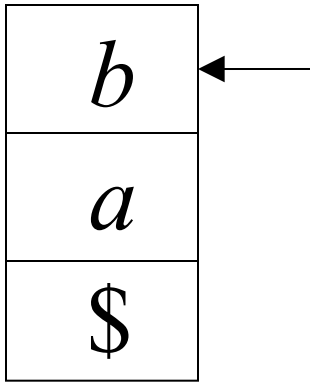
Stack



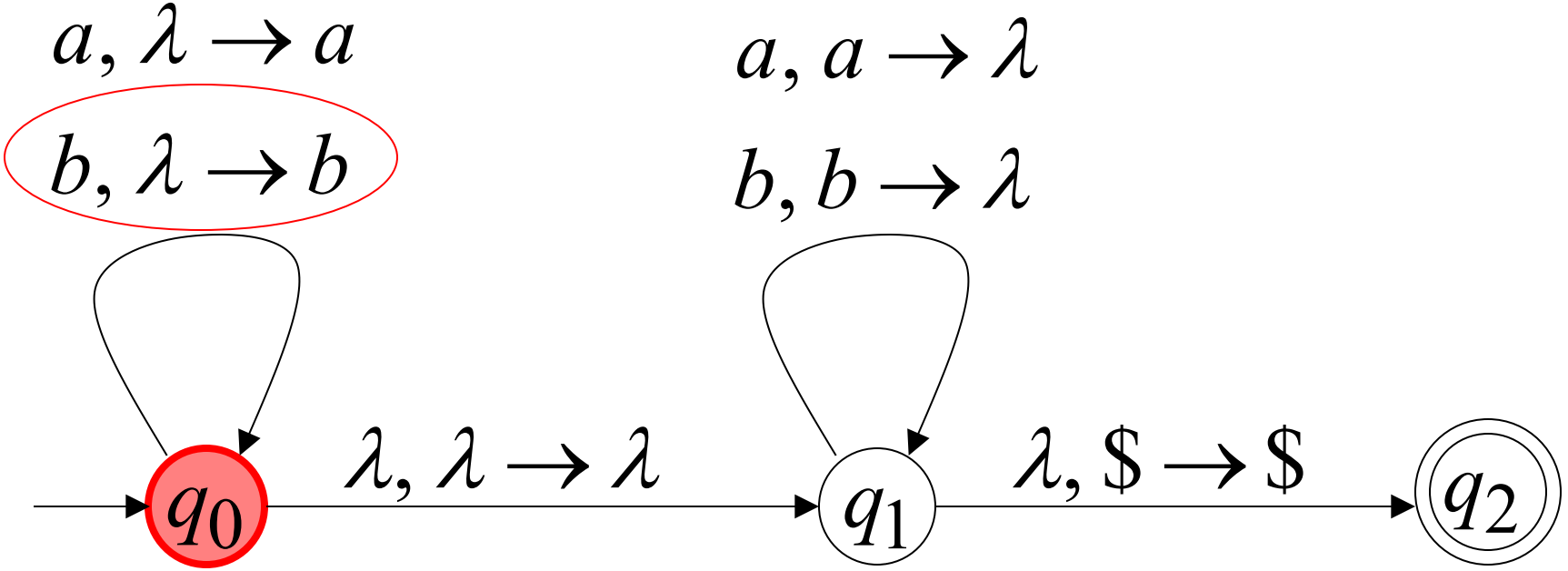
Time 2

Input



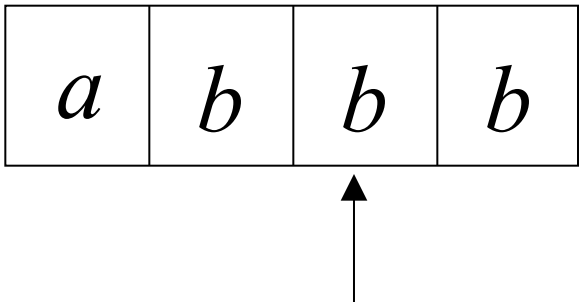


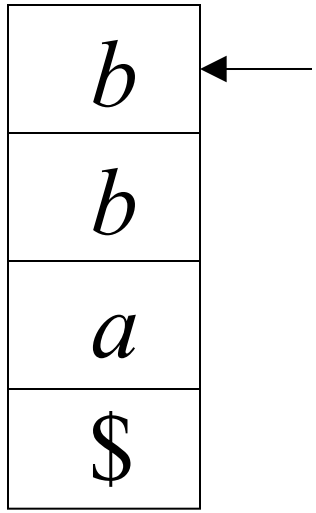
Stack



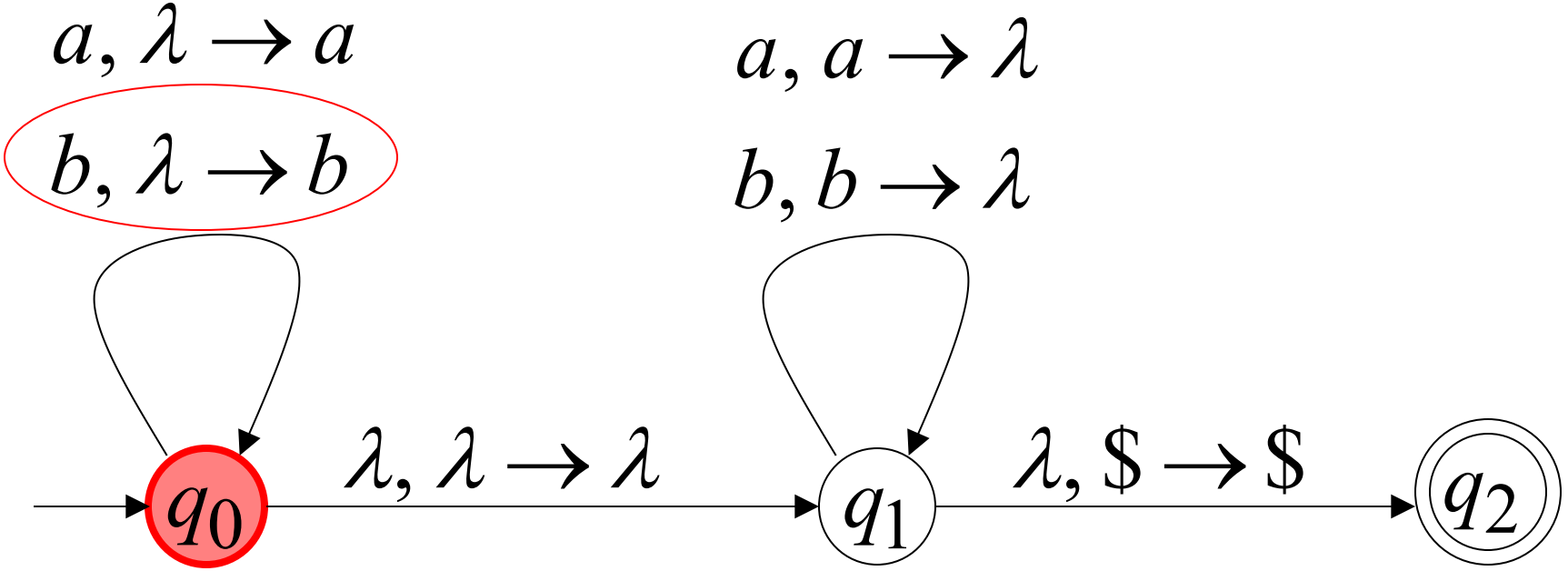
Time 3

Input





Stack



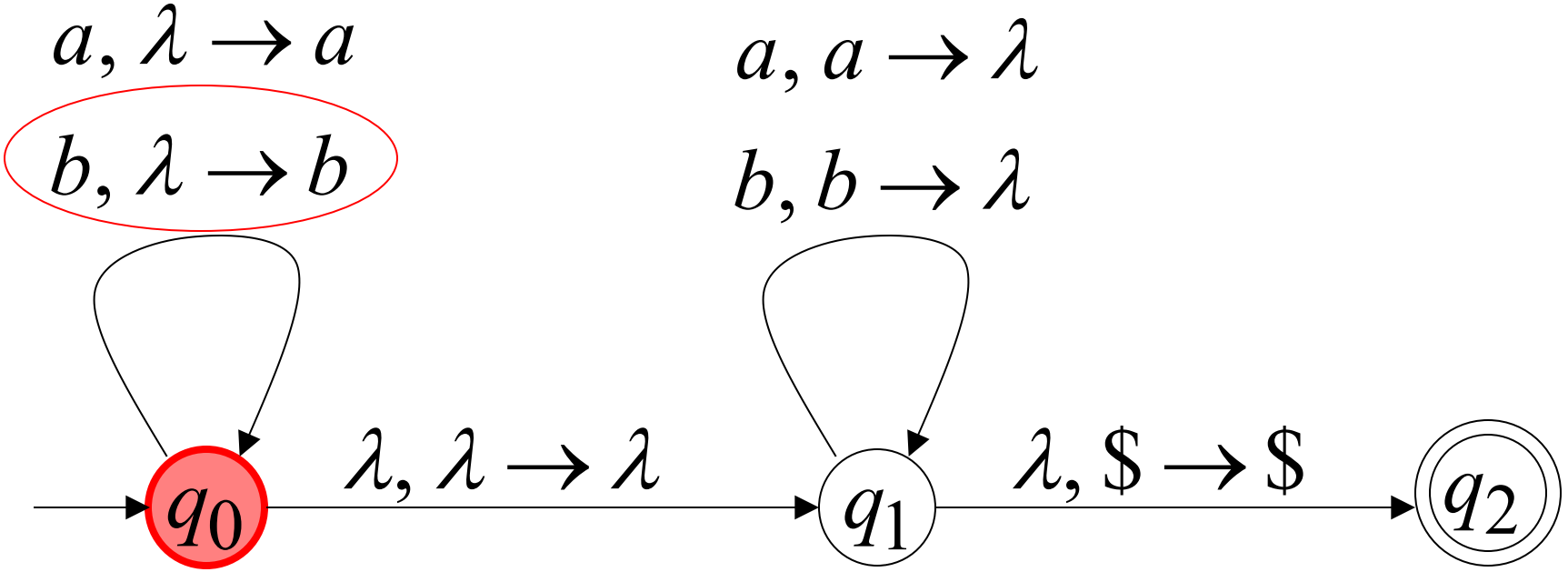
Time 4

Input

a	b	b	b

Stack

b
b
b
a
$\$$



Time 5

Input

a	b	b	b

No final state is reached

b
b
b
a
\$

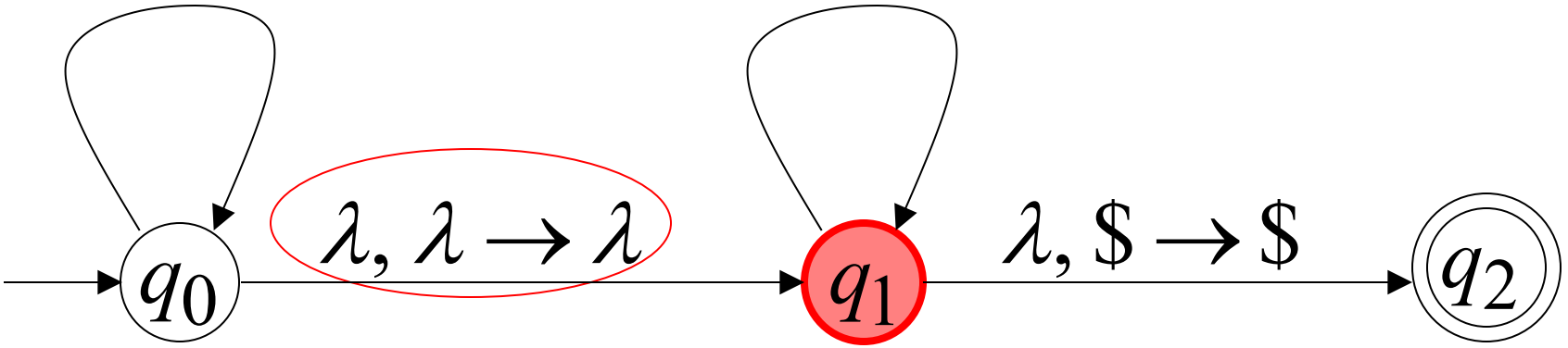
Stack

$a, \lambda \rightarrow a$

$b, \lambda \rightarrow b$

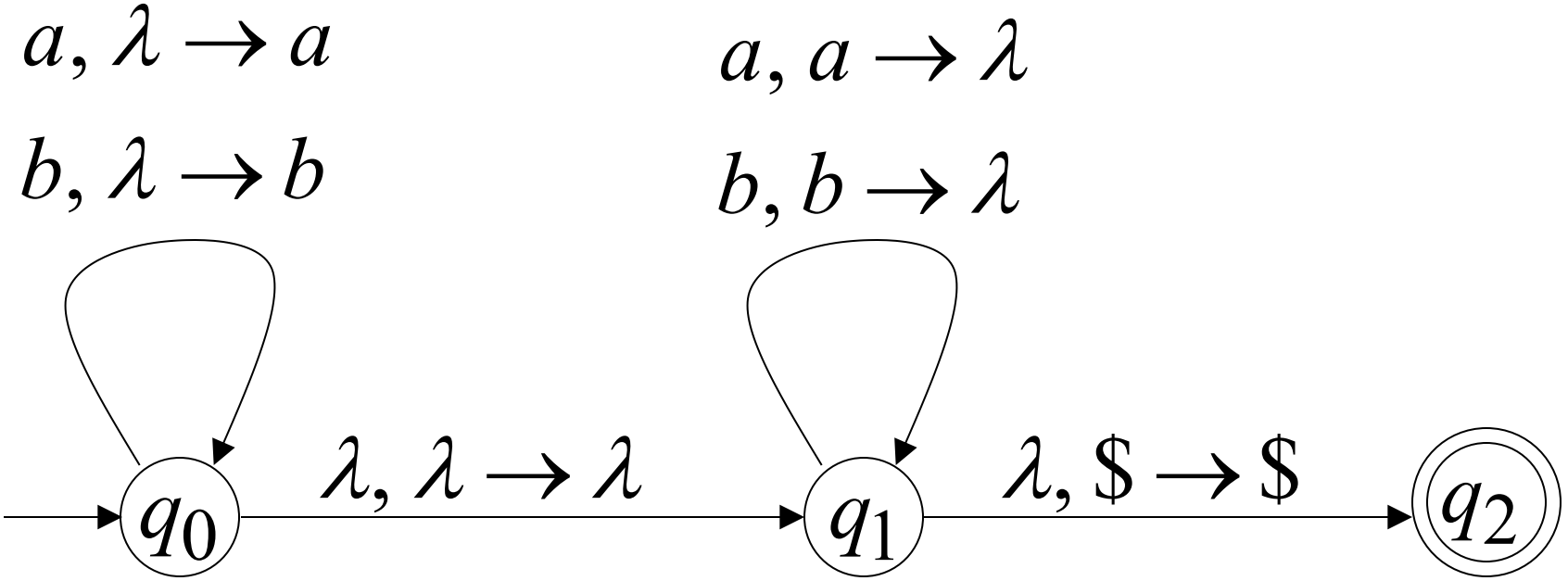
$a, a \rightarrow \lambda$

$b, b \rightarrow \lambda$



There is no computation that accepts string $abbb$

$$abbb \notin L(M)$$



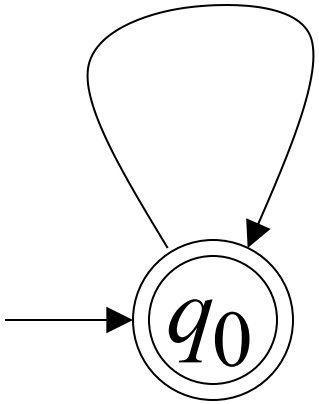
Another PDA example

$$L(M) = \{w \in \{a,b\}^* : \text{in every prefix } v, \quad n_a(v) \geq n_b(v)\}$$

$$a, \lambda \rightarrow a$$

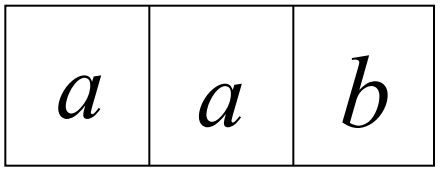
$$b, a \rightarrow \lambda$$

PDA M



Execution Example: Time 0

Input

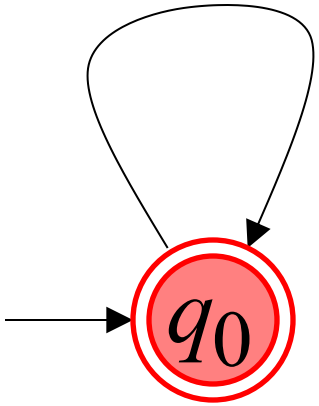


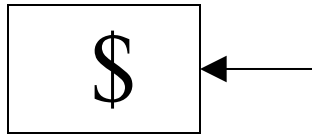


$a, \lambda \rightarrow a$

$b, a \rightarrow \lambda$

$b, \$ \rightarrow \lambda$

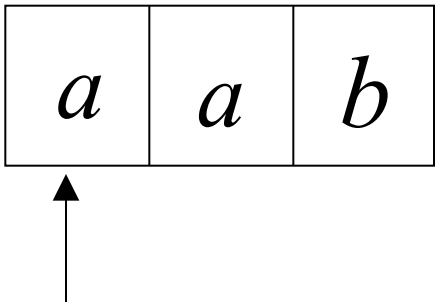


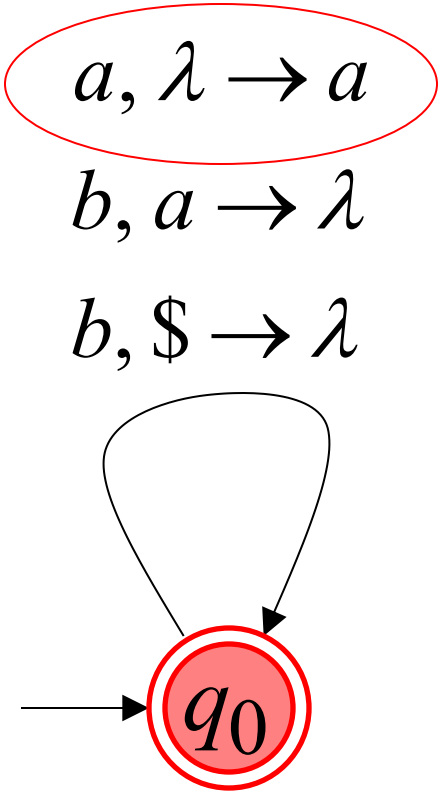


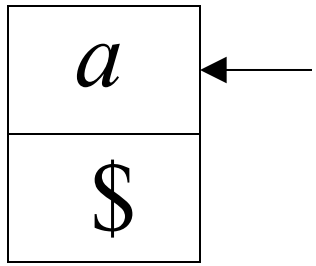
Stack

Time 1

Input



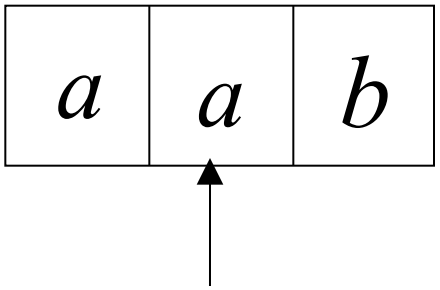


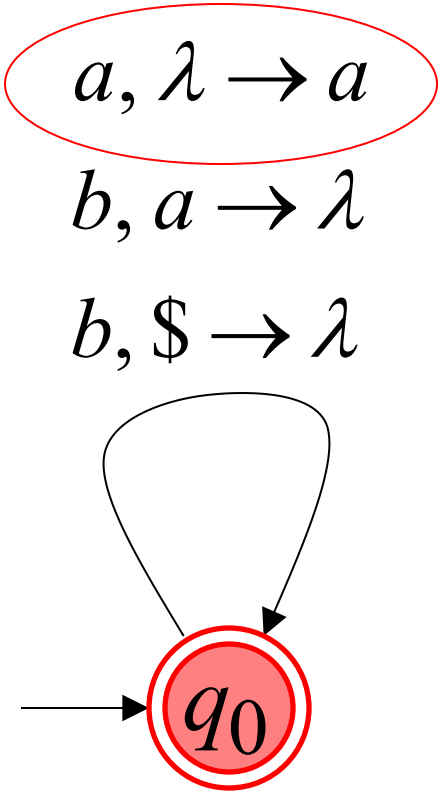


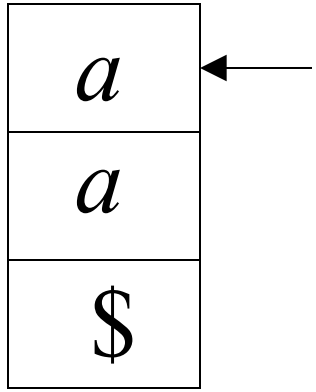
Stack

Time 2

Input



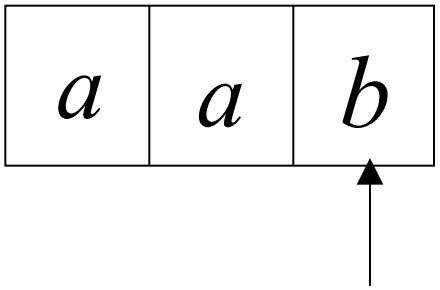




Stack

Time 3

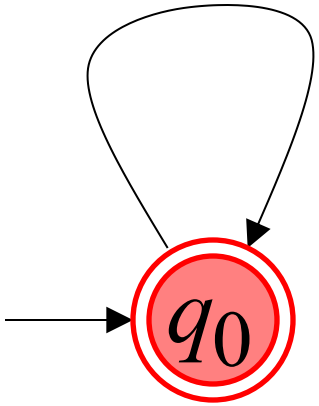
Input

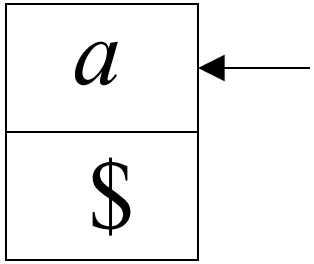


$a, \lambda \rightarrow a$

$b, a \rightarrow \lambda$

$b, \$ \rightarrow \lambda$





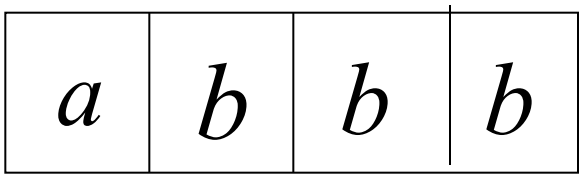
Stack

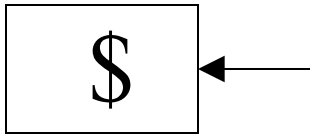
accept

Rejection example:

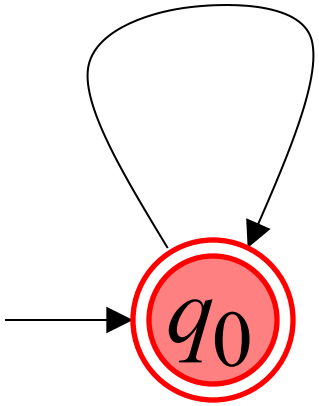
Time 0

Input



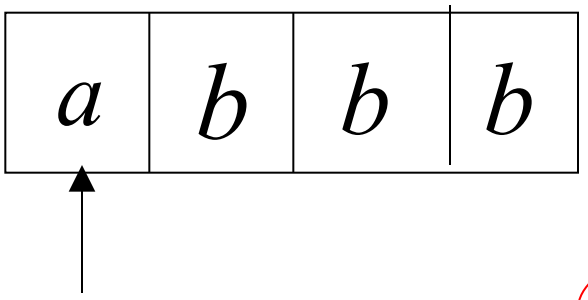


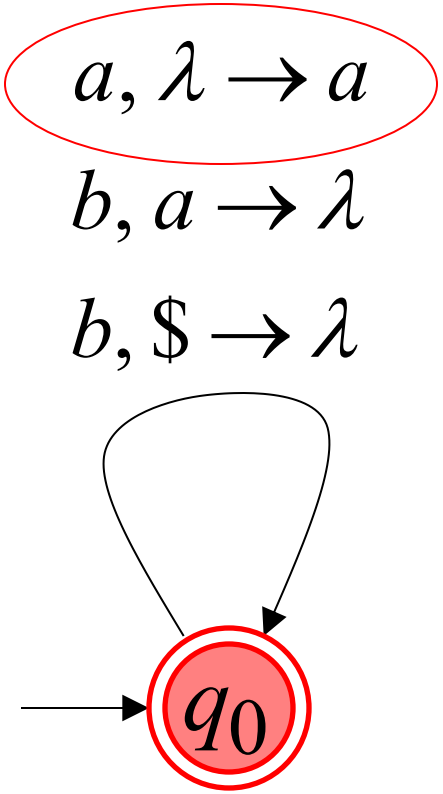
Stack

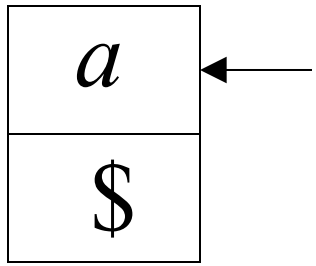


Time 1

Input



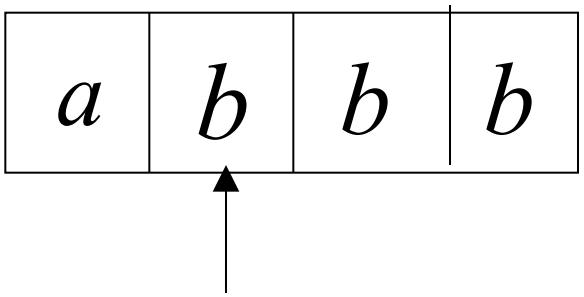




Stack

Time 2

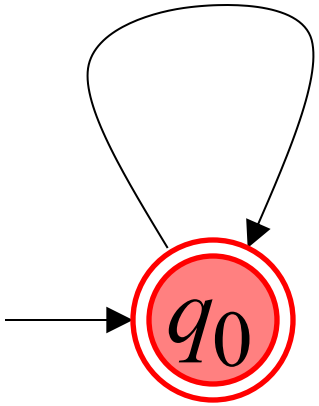
Input

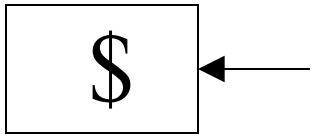


$a, \lambda \rightarrow a$

$b, a \rightarrow \lambda$

$b, \$ \rightarrow \lambda$

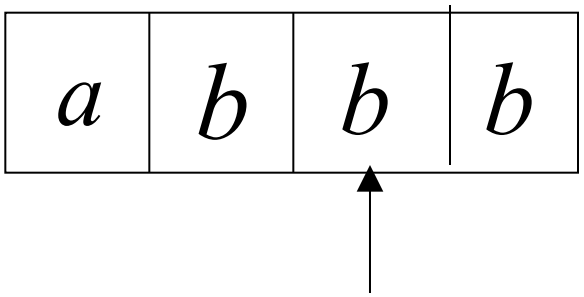




Stack

Time 3

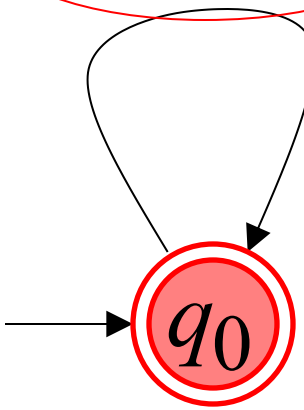
Input

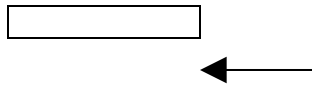


$a, \lambda \rightarrow a$

$b, a \rightarrow \lambda$

$b, \$ \rightarrow \lambda$

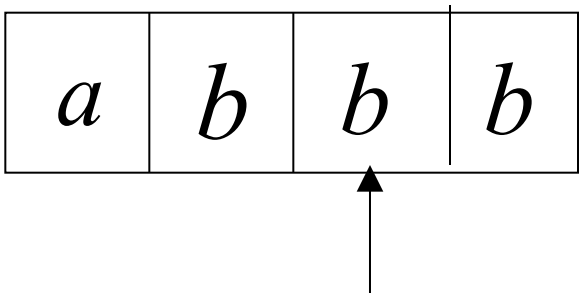




Stack

Time 4

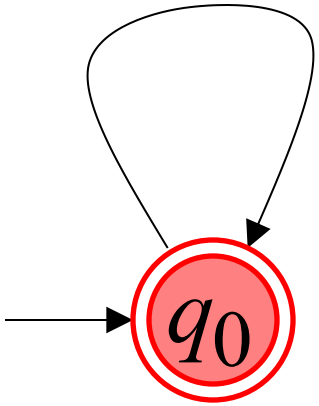
Input

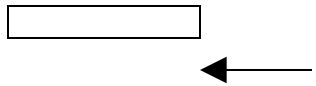


$a, \lambda \rightarrow a$

$b, a \rightarrow \lambda$

$b, \$ \rightarrow \lambda$

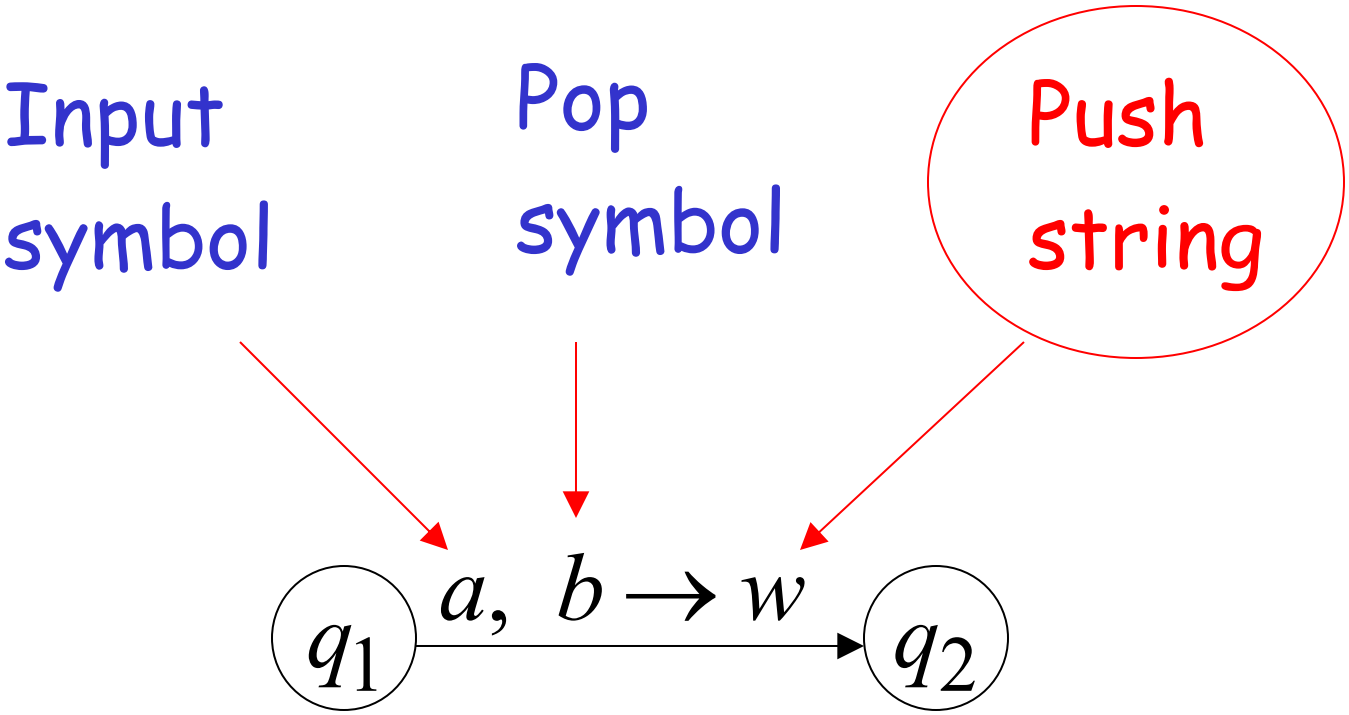




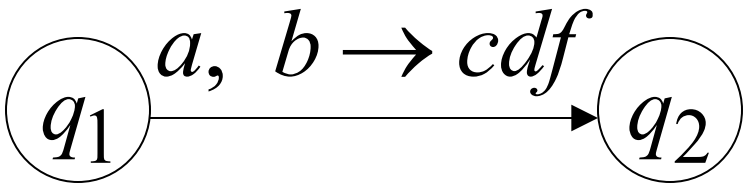
Stack

Halt and Reject

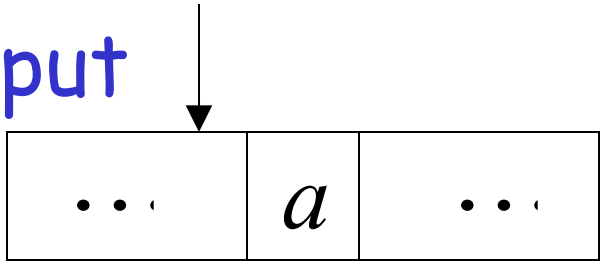
Pushing Strings

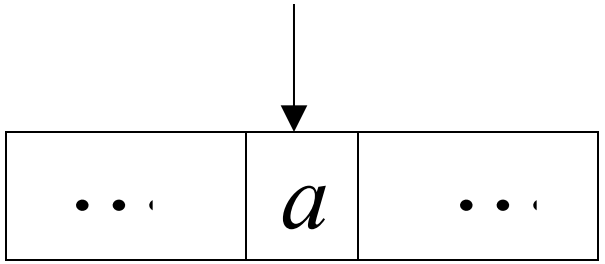


Example:

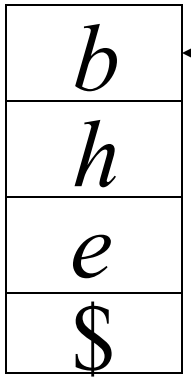


input



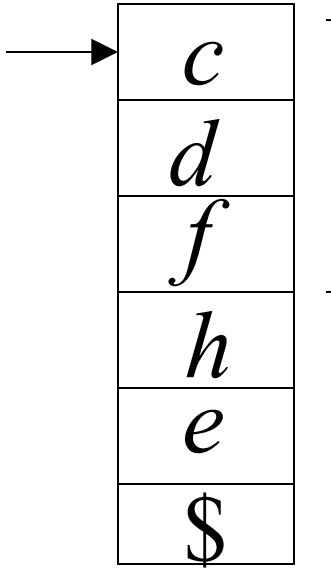


stack



top

Push



pushed string

Another PDA example

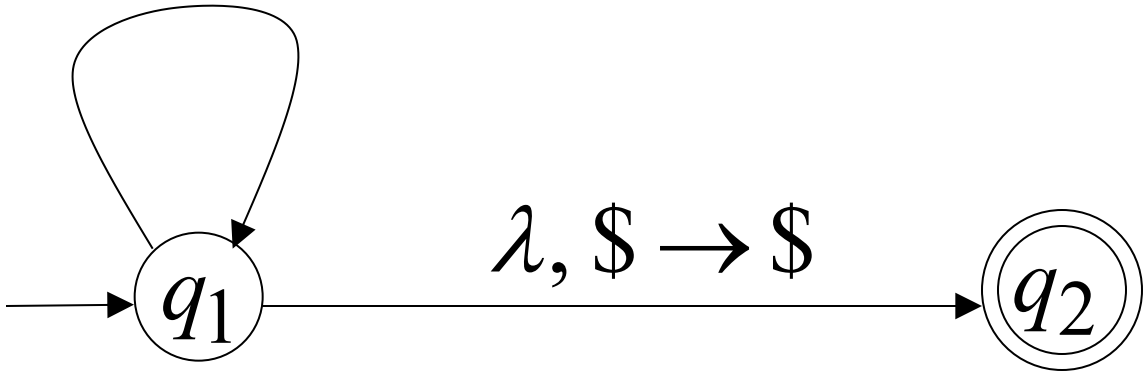
$$L(M) = \{w \in \{a,b\}^* : \quad n_a(w) = n_b(w)\}$$

PDA M

$a, \$ \to 0\$$ $\quad b, \$ \to 1\$$

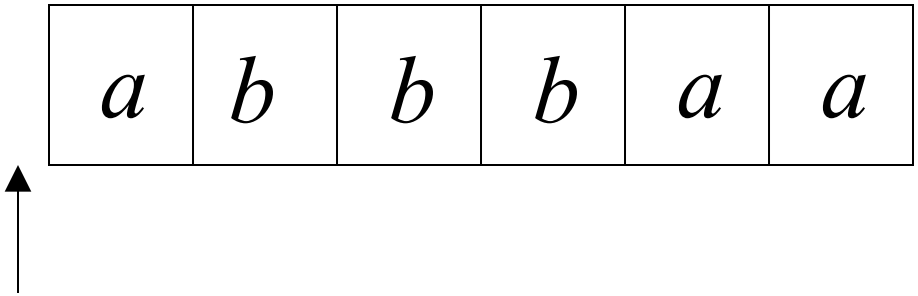
$a, 0 \to 00$ $\quad b, 1 \to 11$

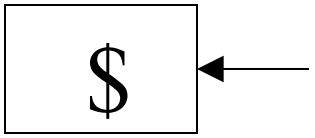
$a, 1 \to \lambda$ $\quad b, 0 \to \lambda$



Execution Example: Time 0

Input





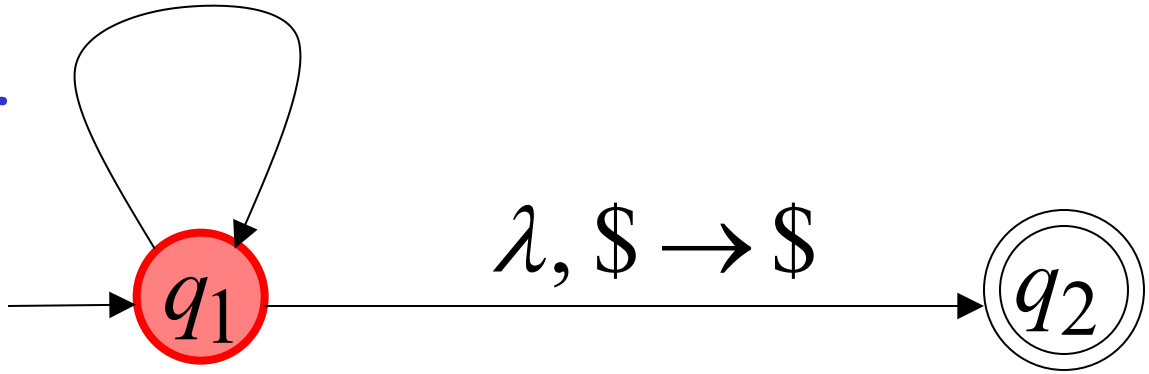
Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

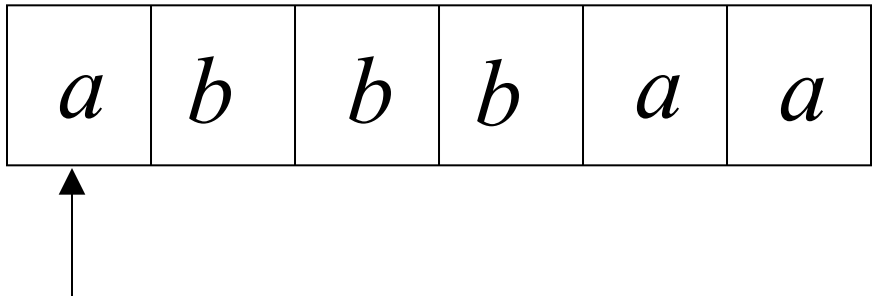
$a, 1 \rightarrow \lambda$ $b, 0 \rightarrow \lambda$

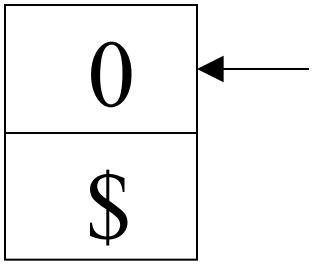
current state



Time 1

Input



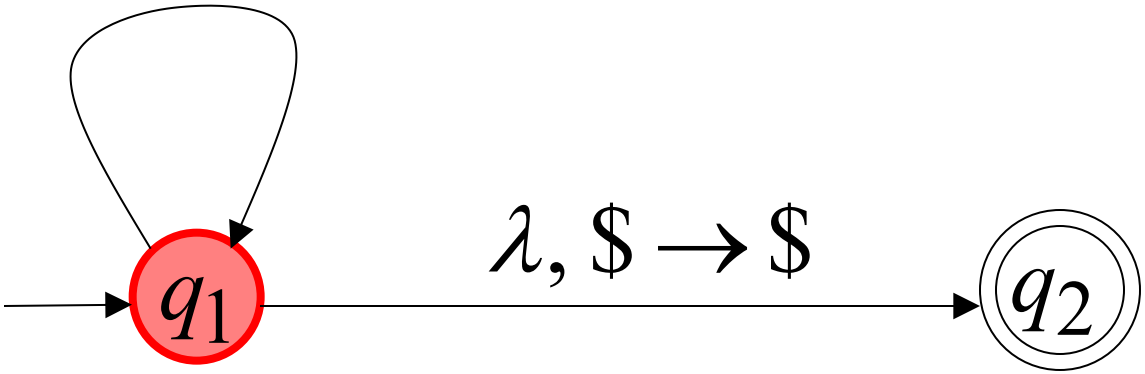


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

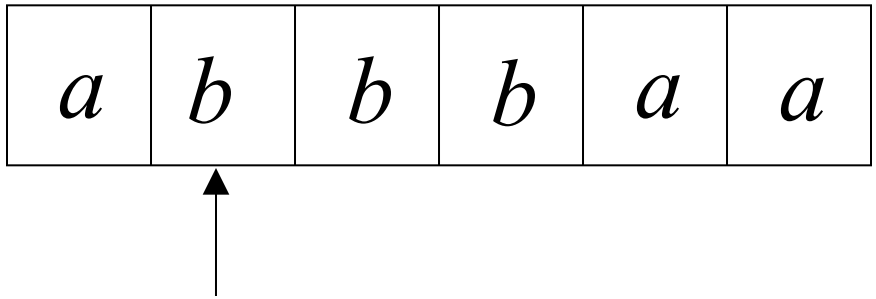
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

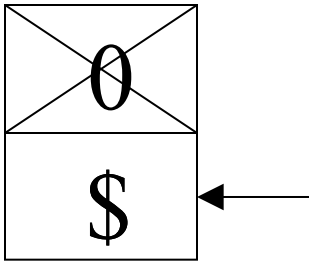
$a, 1 \rightarrow \lambda$ $b, 0 \rightarrow \lambda$



Time 3

Input



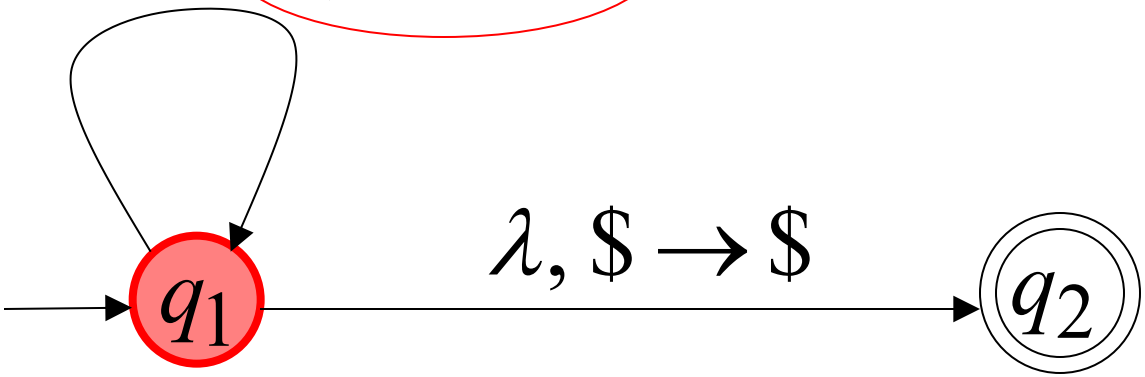


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

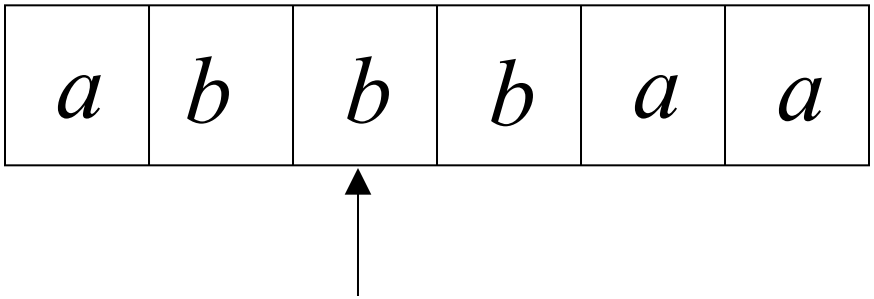
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

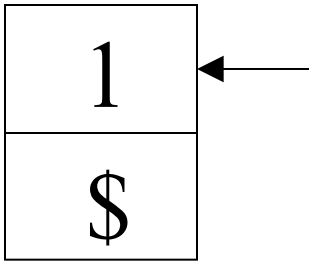
$a, 1 \rightarrow \lambda$ $b, 0 \rightarrow \lambda$



Time 4

Input



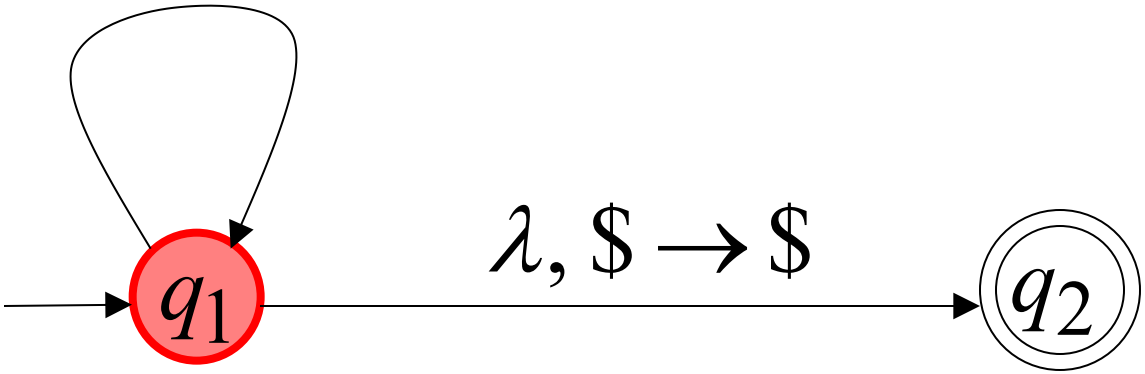


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

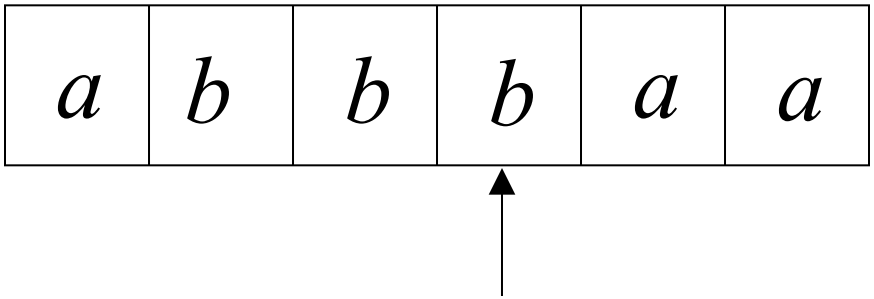
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

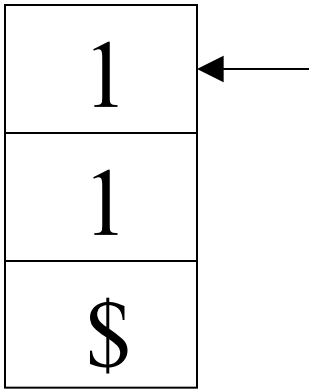
$a, 1 \rightarrow \lambda$ $b, 0 \rightarrow \lambda$



Time 5

Input



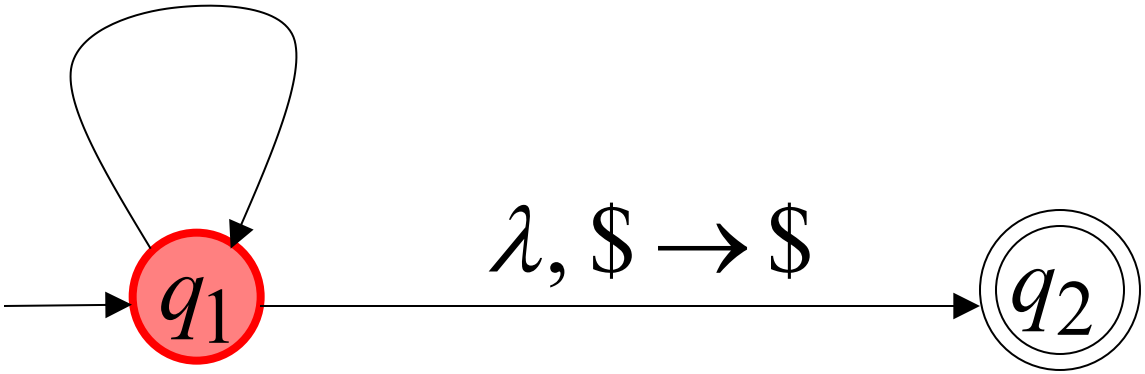


Stack

$$a, \$ \rightarrow 0\$ \qquad b, \$ \rightarrow 1\$$$

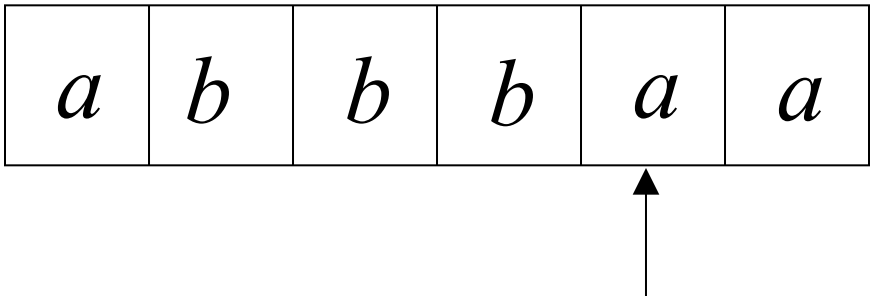
$$a, 0 \rightarrow 00 \qquad b, 1 \rightarrow 11$$

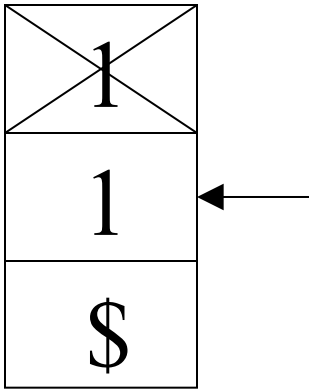
$$a, 1 \rightarrow \lambda \qquad b, 0 \rightarrow \lambda$$



Time 6

Input



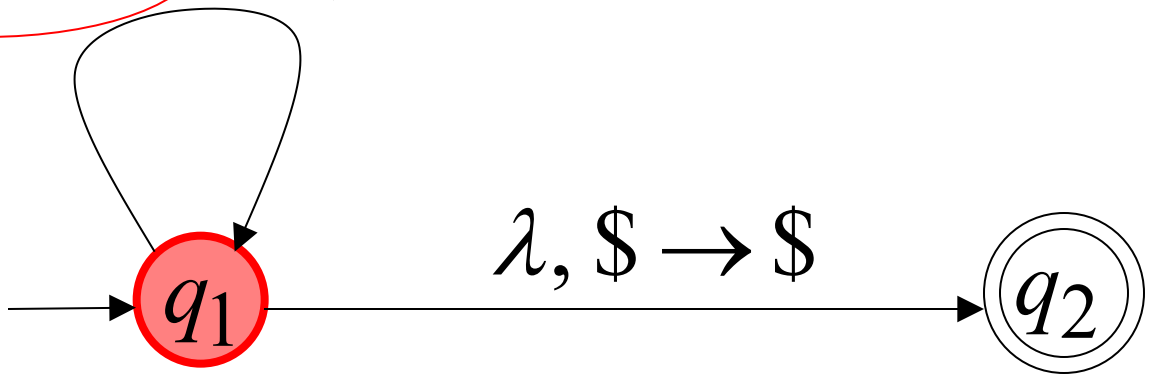


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

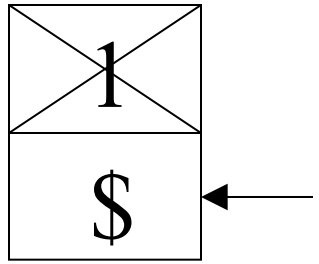
$a, 1 \rightarrow \lambda$ $b, 0 \rightarrow \lambda$



Time 7

Input

a	b	b	b	a	a

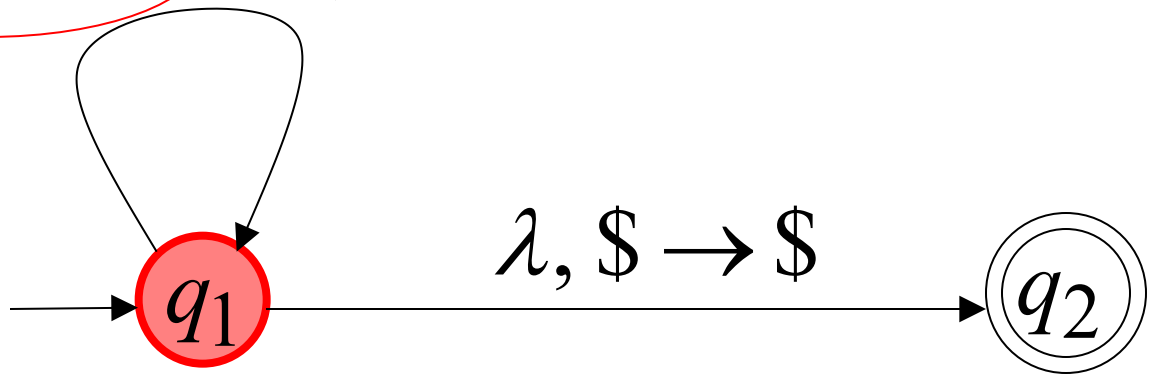


Stack

$a, \$ \rightarrow 0\$$ $b, \$ \rightarrow 1\$$

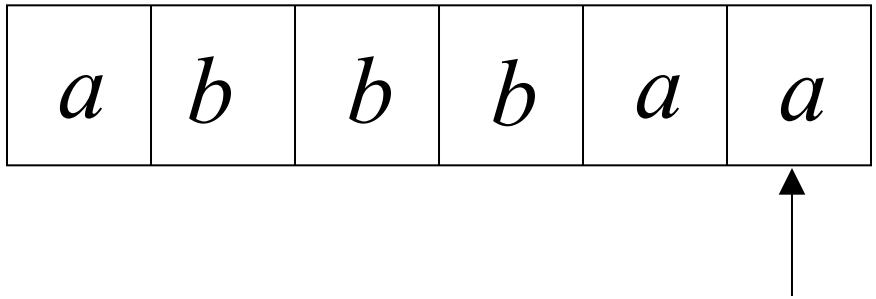
$a, 0 \rightarrow 00$ $b, 1 \rightarrow 11$

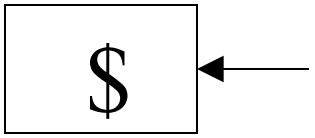
$a, 1 \rightarrow \lambda$ $b, 0 \rightarrow \lambda$



Time 8

Input



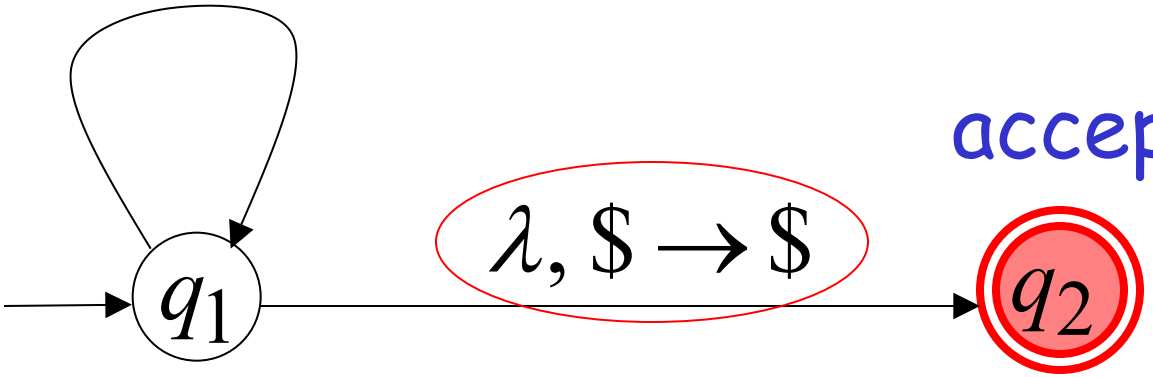


Stack

$$a, \$ \rightarrow 0\$ \qquad b, \$ \rightarrow 1\$$$

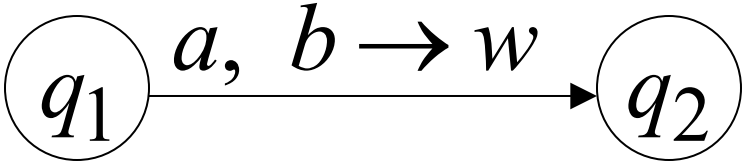
$$a, 0 \rightarrow 00 \qquad b, 1 \rightarrow 11$$

$$a, 1 \rightarrow \lambda \qquad b, 0 \rightarrow \lambda$$



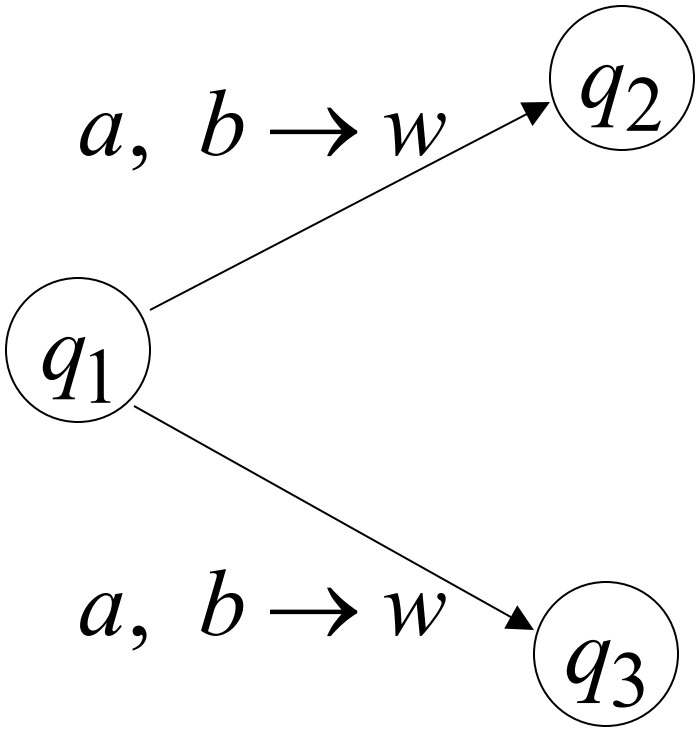
accept

Formalities for PDAs



Transition function:

$$\delta(q_1, a, b) = \{(q_2, w)\}$$



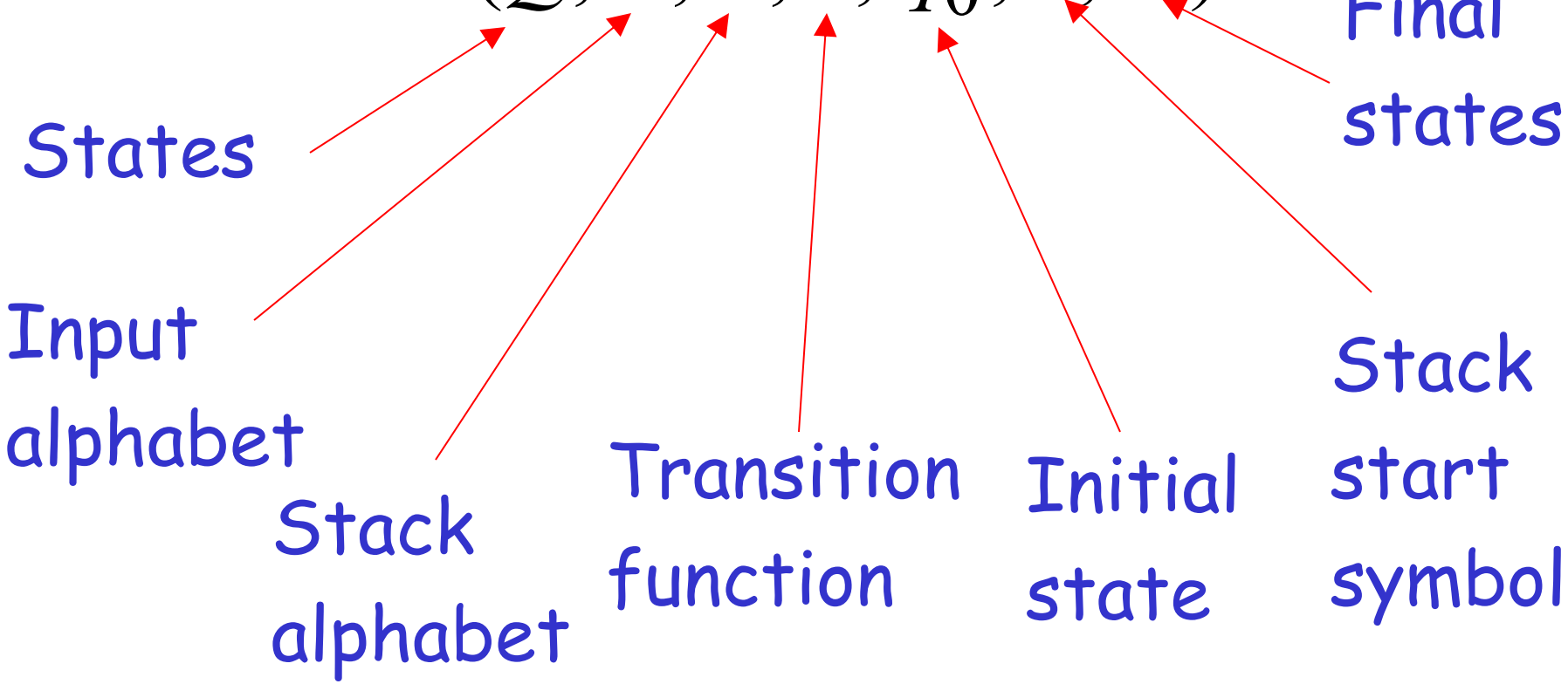
Transition function:

$$\delta(q_1, a, b) = \{(q_2, w),\ (q_3, w)\}$$

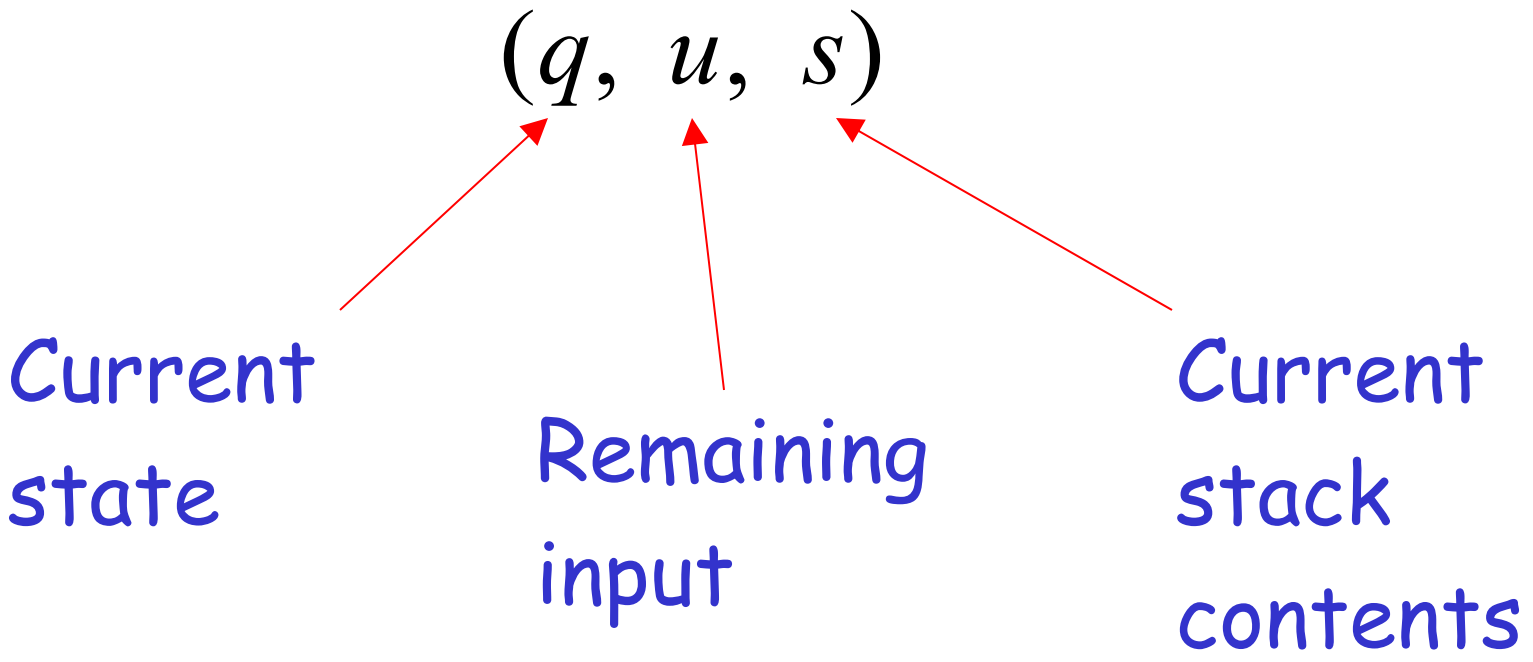
Formal Definition

Pushdown Automaton (PDA)

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$$



Instantaneous Description

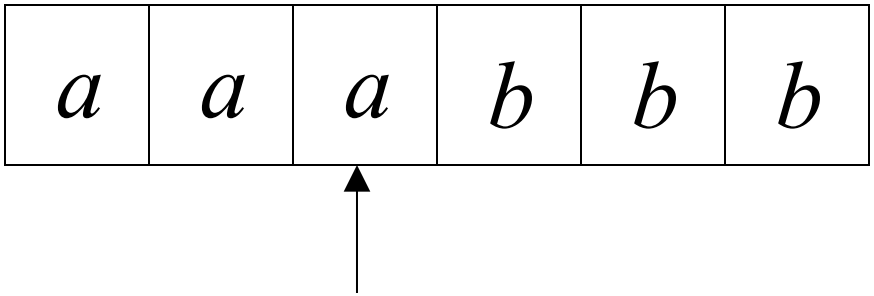


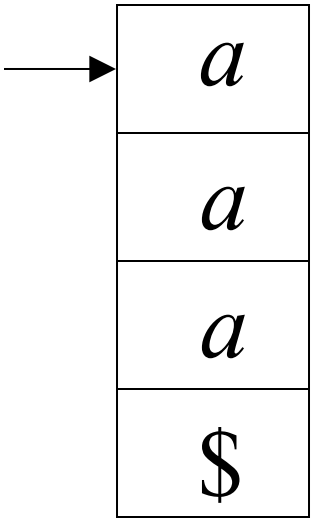
Example: **Instantaneous Description**

$$(q_1, bbb, aaa\$)$$

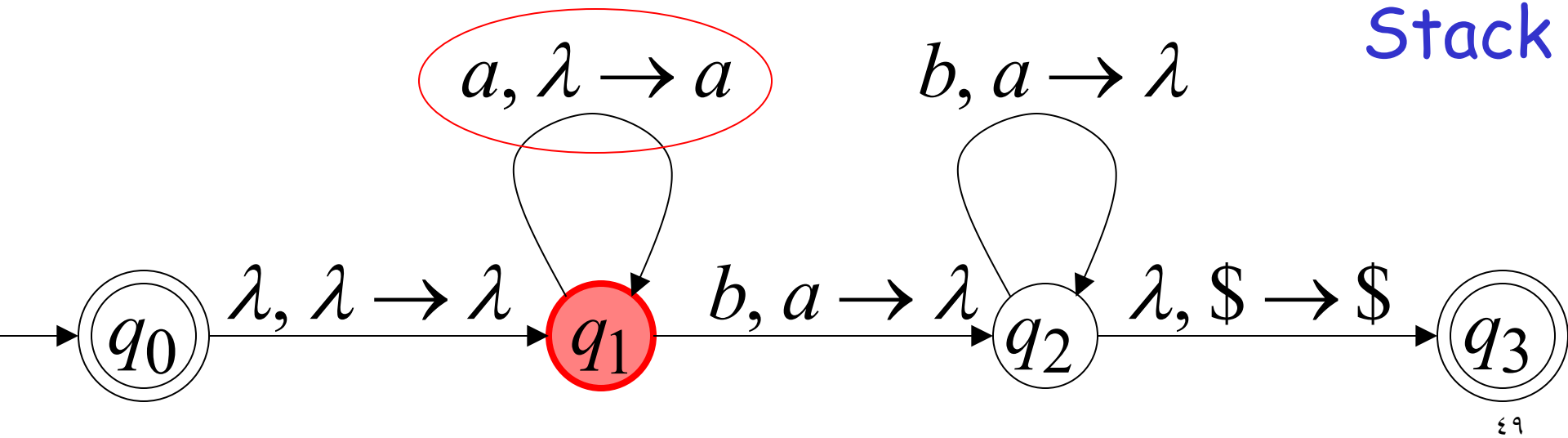
Time 4:

Input





Stack

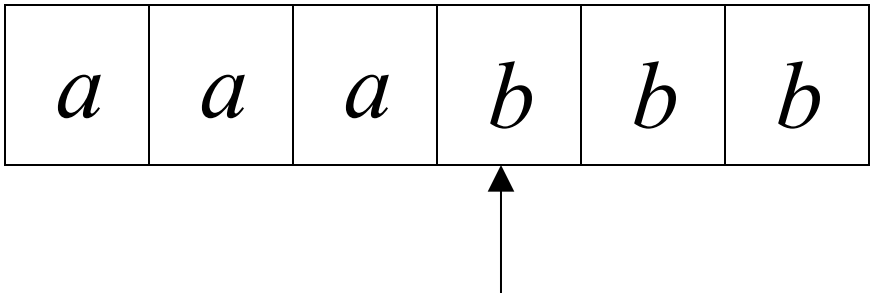


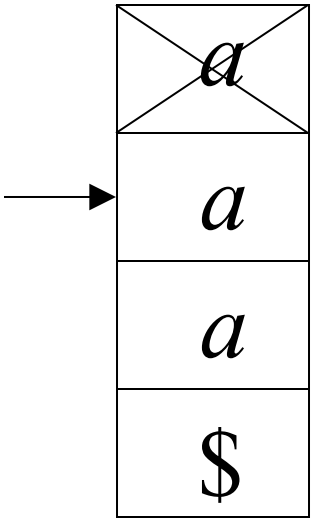
Example: **Instantaneous Description**

$$(q_2, bb, aa\$)$$

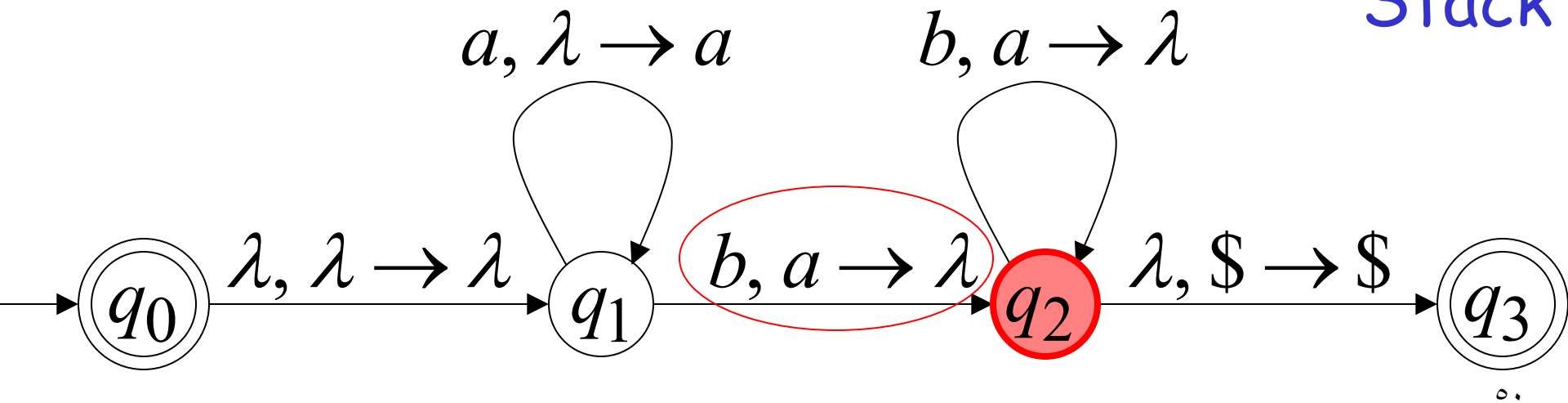
Time 5:

Input





Stack



We write:

$$(q_1, bbb, aaa\$) \succ (q_2, bb, aa\$)$$

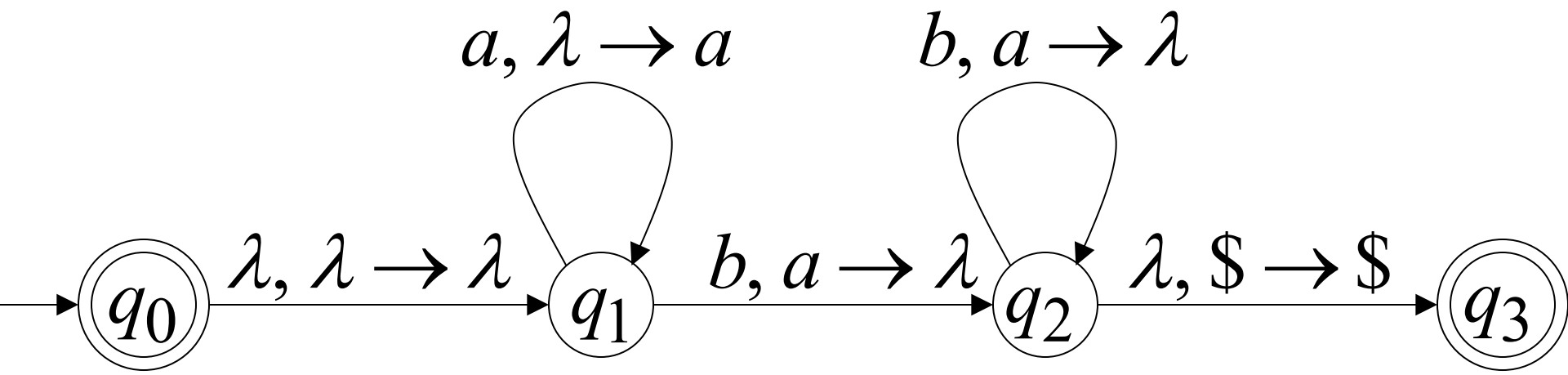
Time 4 Time 5

A computation:

$$(q_0, aaabbb, \$) \succ (q_1, aaabbb, \$) \succ$$

$$(q_1, aabbb, a\$) \succ (q_1, abbb, aa\$) \succ (q_1, bbb, aaa\$) \succ$$

$$(q_2, bb, aa\$) \succ (q_2, b, a\$) \succ (q_2, \lambda, \$) \succ (q_3, \lambda, \$)$$



$$(q_0, aaabbb, \$) \succ (q_1, aaabbb, \$) \succ$$

$$(q_1, aabbb, a\$) \succ (q_1, abbb, aa\$) \succ (q_1, bbb, aaa\$) \succ$$

$$(q_2, bb, aa\$) \succ (q_2, b, a\$) \succ (q_2, \lambda, \$) \succ (q_3, \lambda, \$)$$

For convenience we write:

$$(q_0, aaabbb, \$) \overset{*}{\succ} (q_3, \lambda, \$)$$

Formal Definition

Language $L(M)$ of PDA M :

$$L(M) = \{w: \ (q_0, w, s) \overset{*}{\succ} (q_f, \lambda, s')\}$$

Initial state

Final state

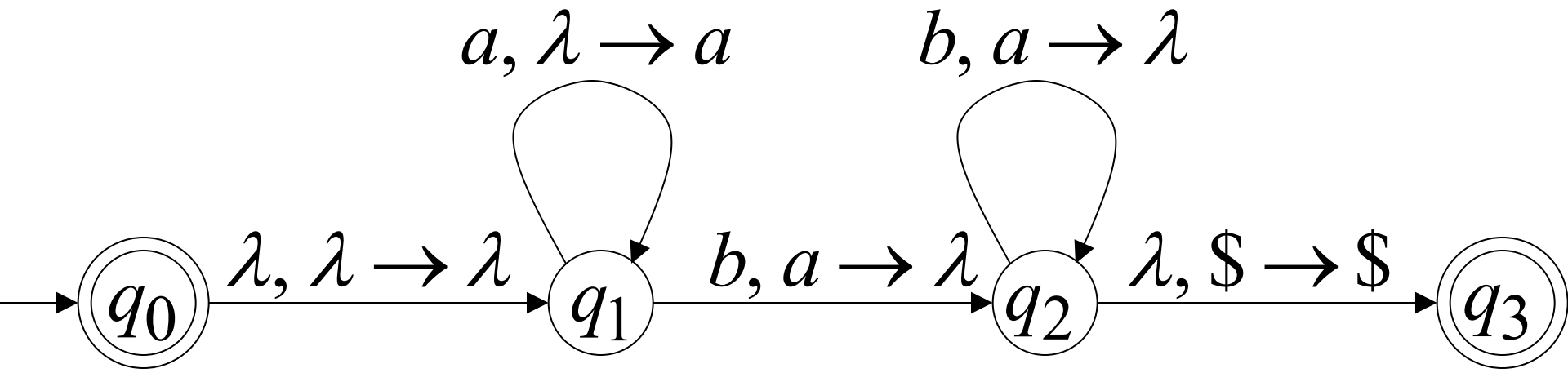
Example:

$$(q_0, aaabbb, \$) \overset{*}{\succ} (q_3, \lambda, \$)$$



$$aaabbb \in L(M)$$

PDA M:

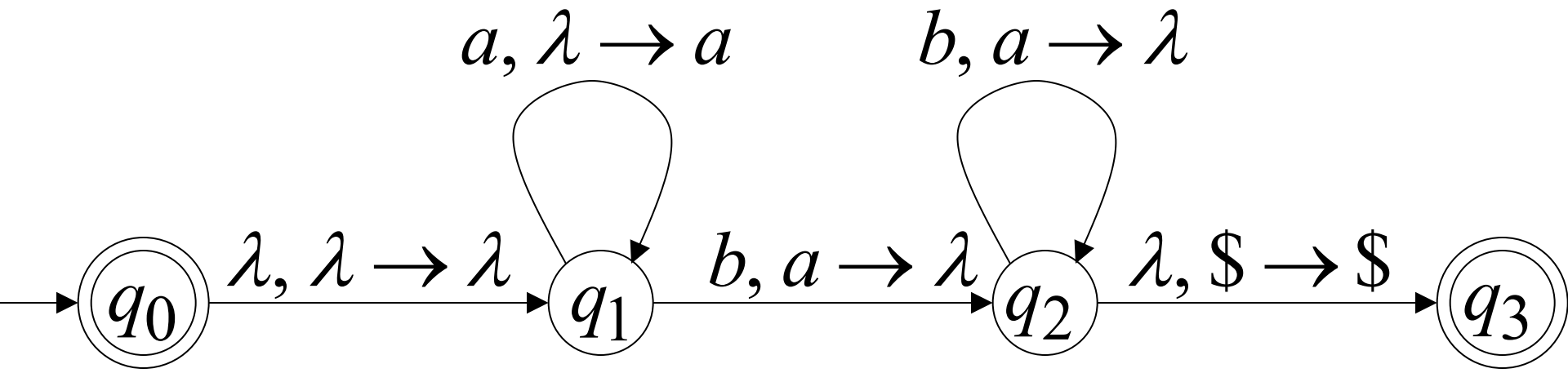


$$(q_0, a^n b^n, \$) \overset{*}{\succ} (q_3, \lambda, \$)$$



$$a^n b^n \in L(M)$$

PDA M :



Therefore: $L(M) = \{a^n b^n : n \geq 0\}$

PDA M :

